



US 20250283659A1

(19) **United States**(12) **Patent Application Publication**
Alexander et al.(10) **Pub. No.: US 2025/0283659 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **PORTABLE COOLER WITH ACTIVE TEMPERATURE CONTROL**

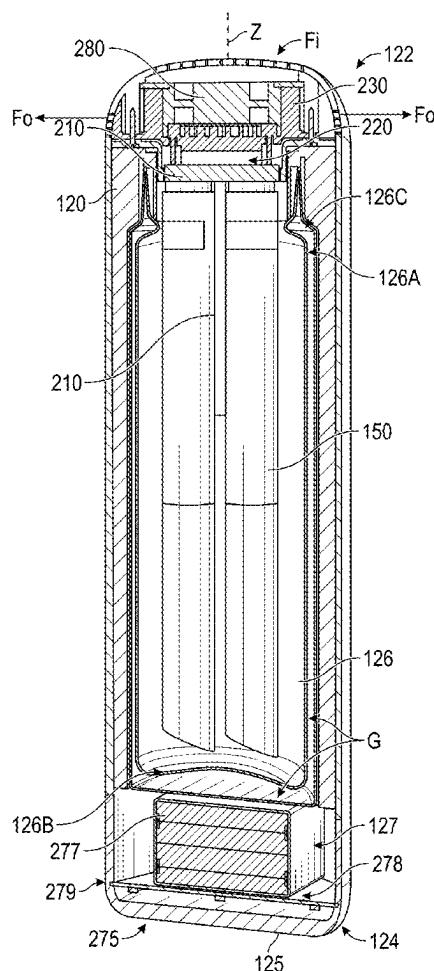
(60) Provisional application No. 62/791,225, filed on Jan. 11, 2019, provisional application No. 62/827,636, filed on Apr. 1, 2019.

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F25D 31/00 (2006.01)
F25B 21/02 (2006.01)
F25B 21/04 (2006.01)
(52) **U.S. Cl.**
CPC **F25D 31/005** (2013.01); **F25B 21/02** (2013.01); **F25B 21/04** (2013.01); **F25D 31/006** (2013.01); **F25B 2321/023** (2013.01); **F25B 2321/0251** (2013.01); **F25D 2331/803** (2013.01); **F25D 2331/805** (2013.01); **F25D 2400/12** (2013.01); **F25D 2700/12** (2013.01)(21) Appl. No.: **19/215,188**(22) Filed: **May 21, 2025****Related U.S. Application Data**

(63) Continuation of application No. 17/301,932, filed on Apr. 19, 2021, which is a continuation of application No. 16/736,558, filed on Jan. 7, 2020, now Pat. No. 10,989,466.

(57) **ABSTRACT**

A portable cooler container with active temperature control system is provided. The active temperature control system is operated to heat or cool a chamber of a vessel to approach a temperature set point suitable for a medication stored in the cooler container.



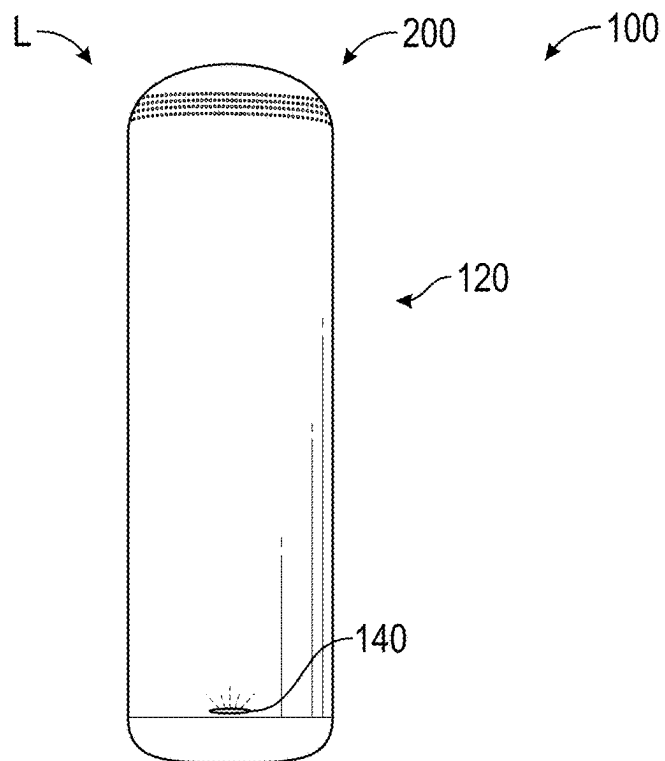


FIG. 1

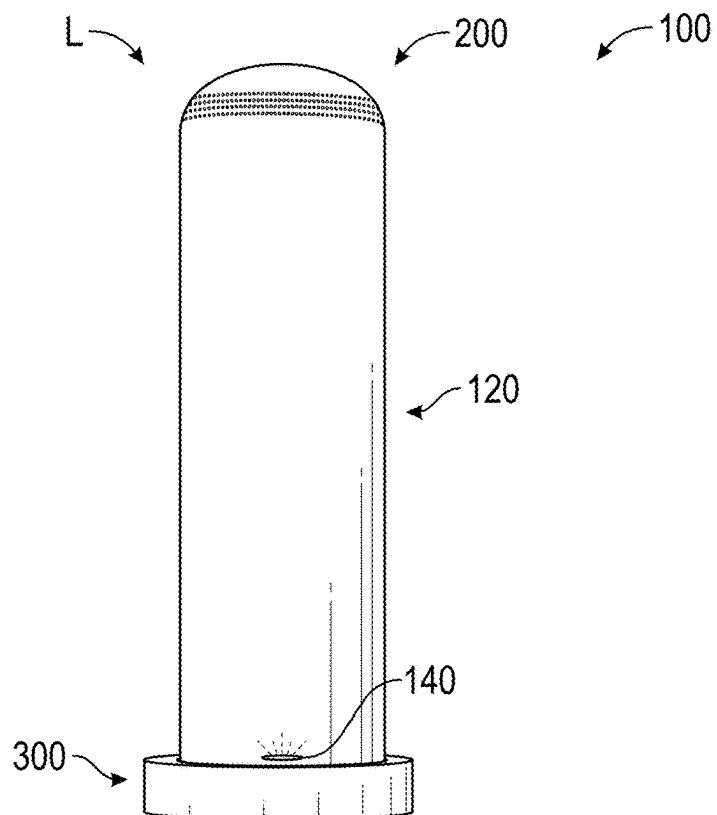


FIG. 2

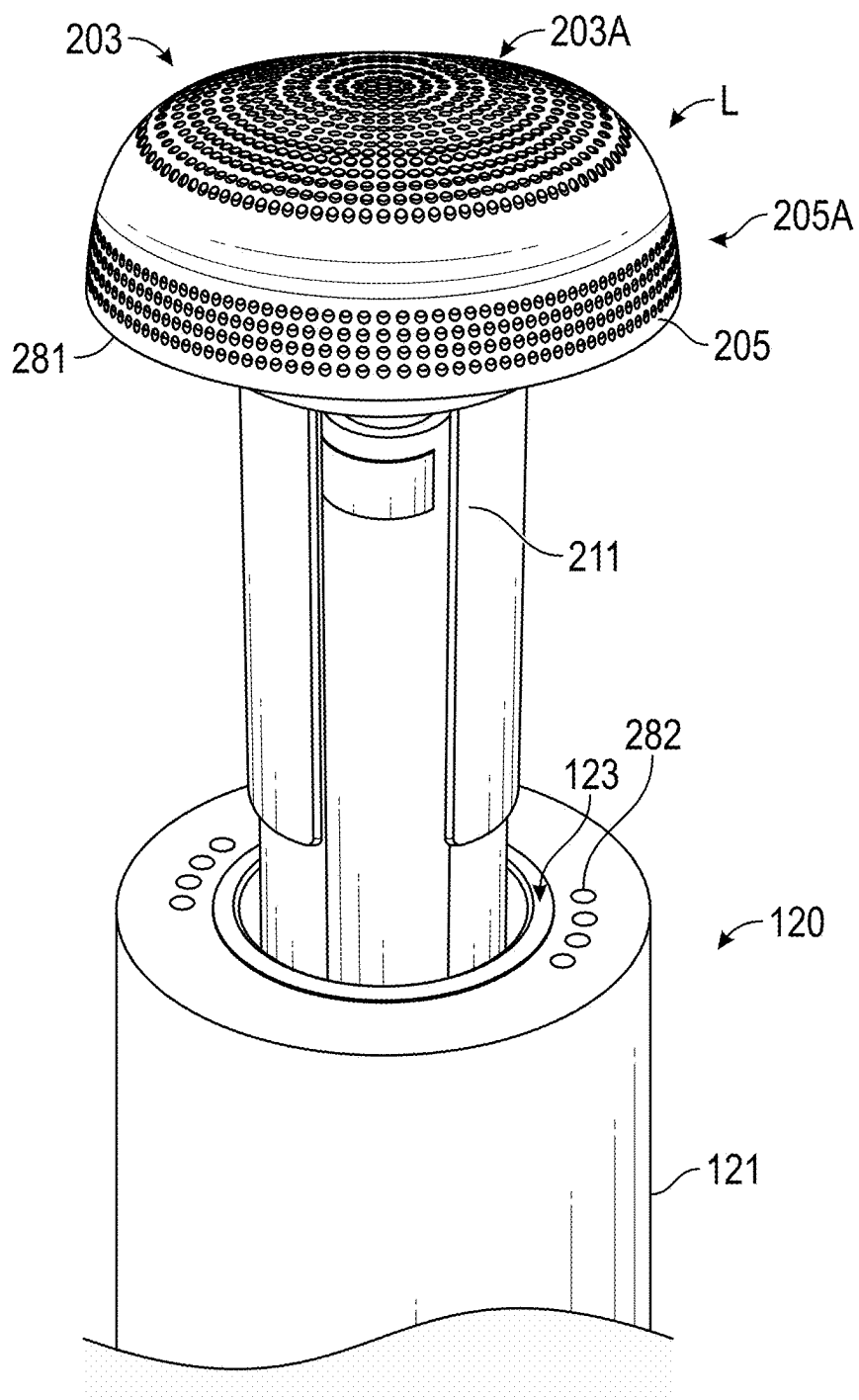


FIG. 3

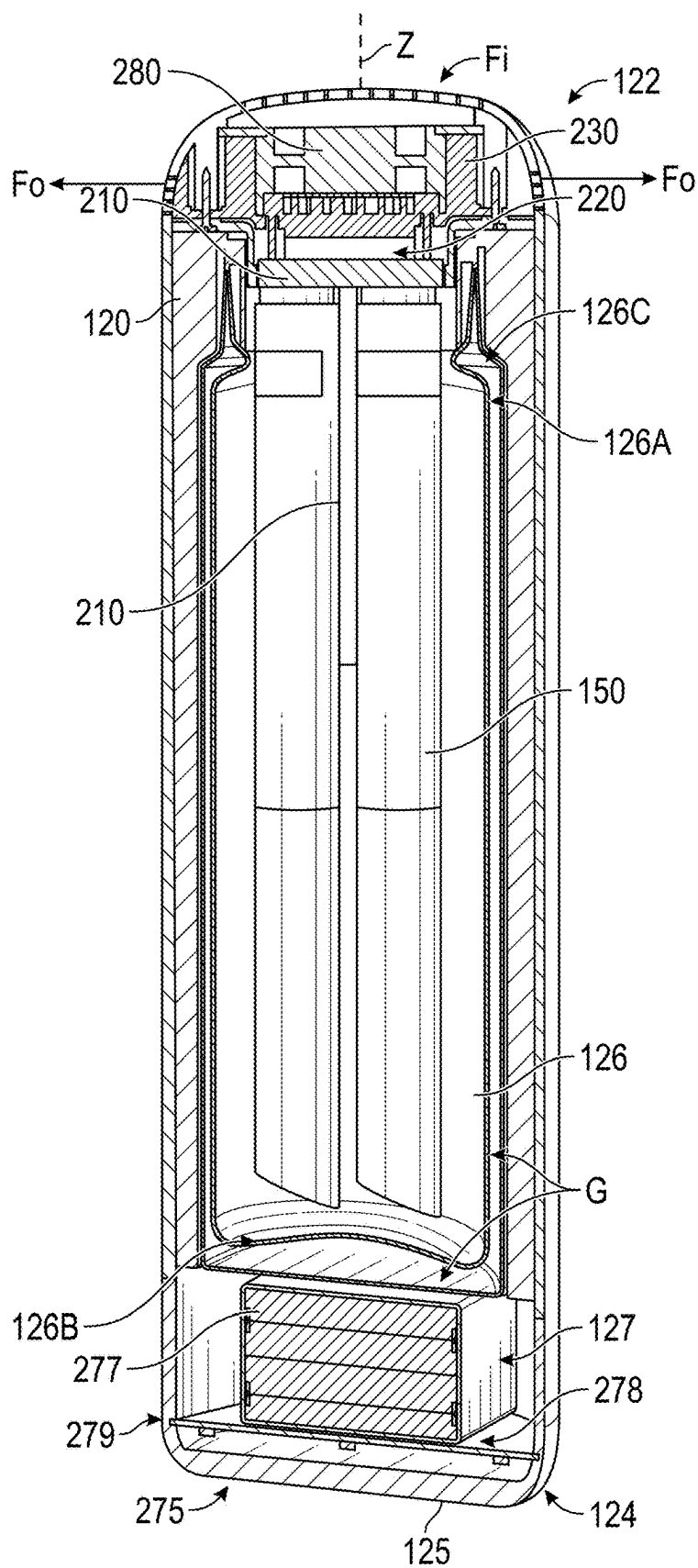
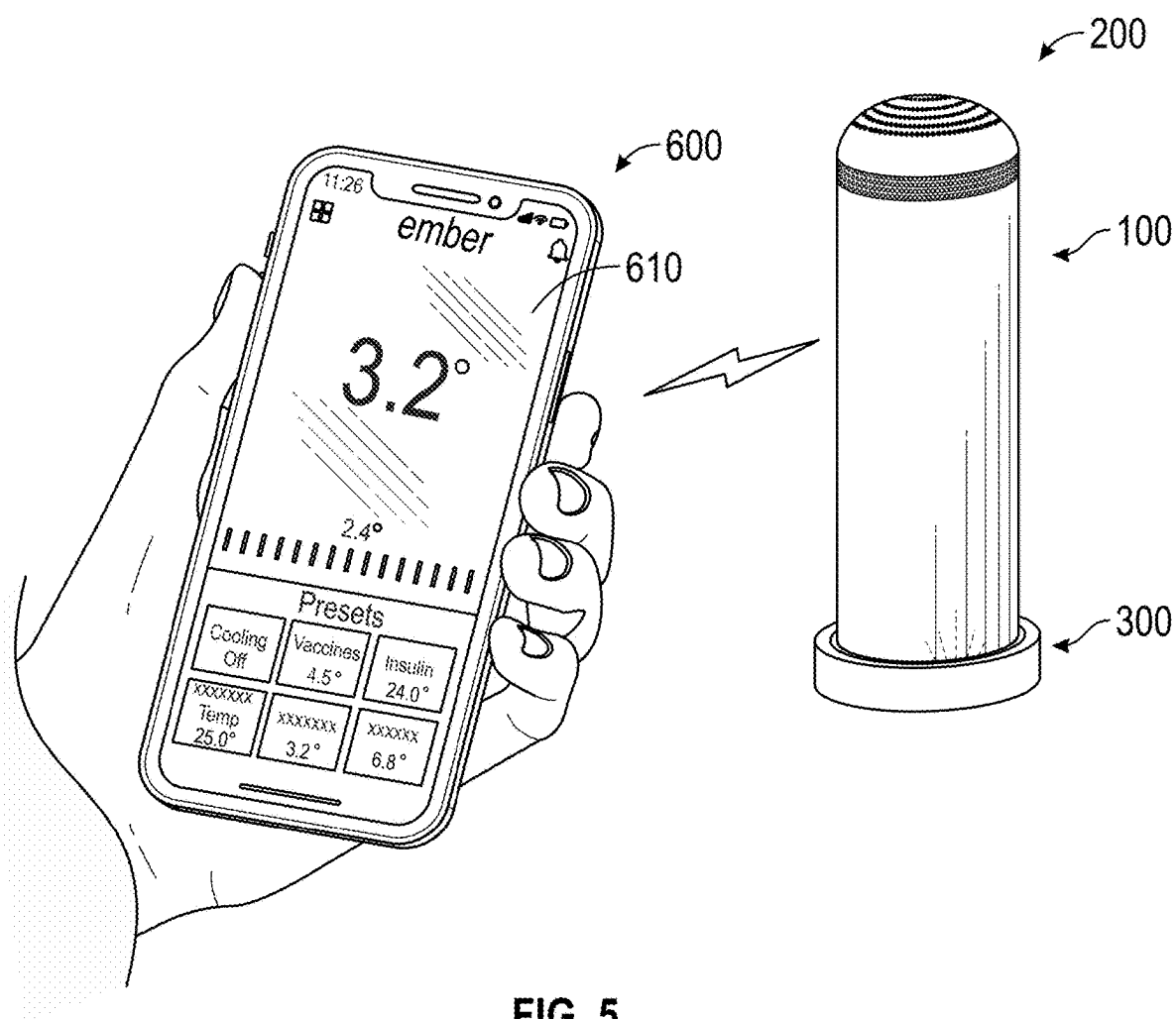


FIG. 4



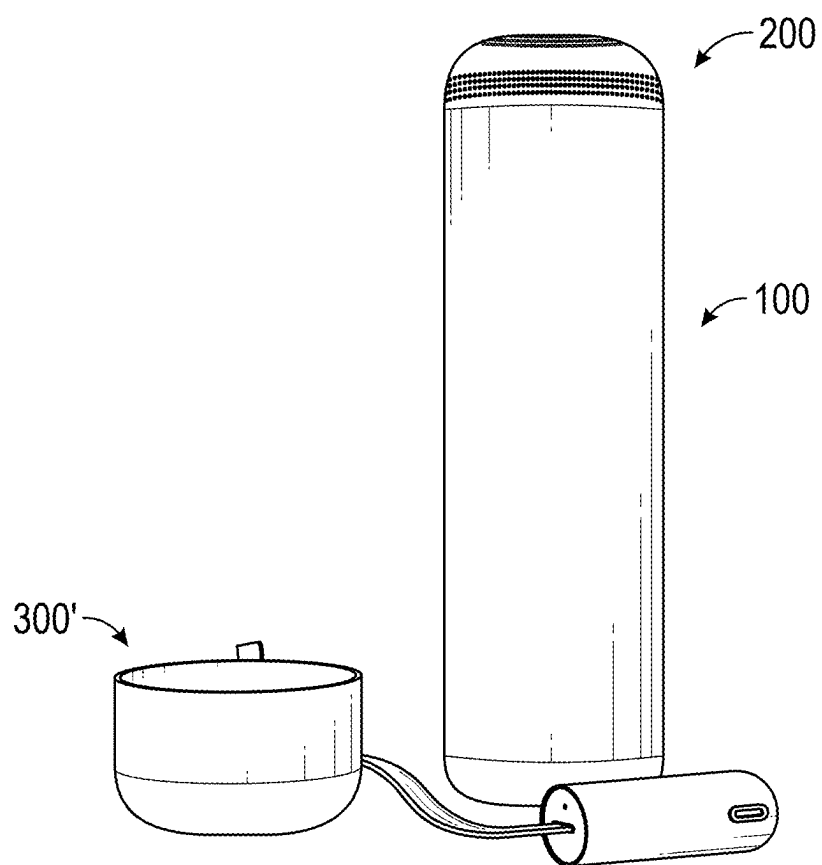


FIG. 6

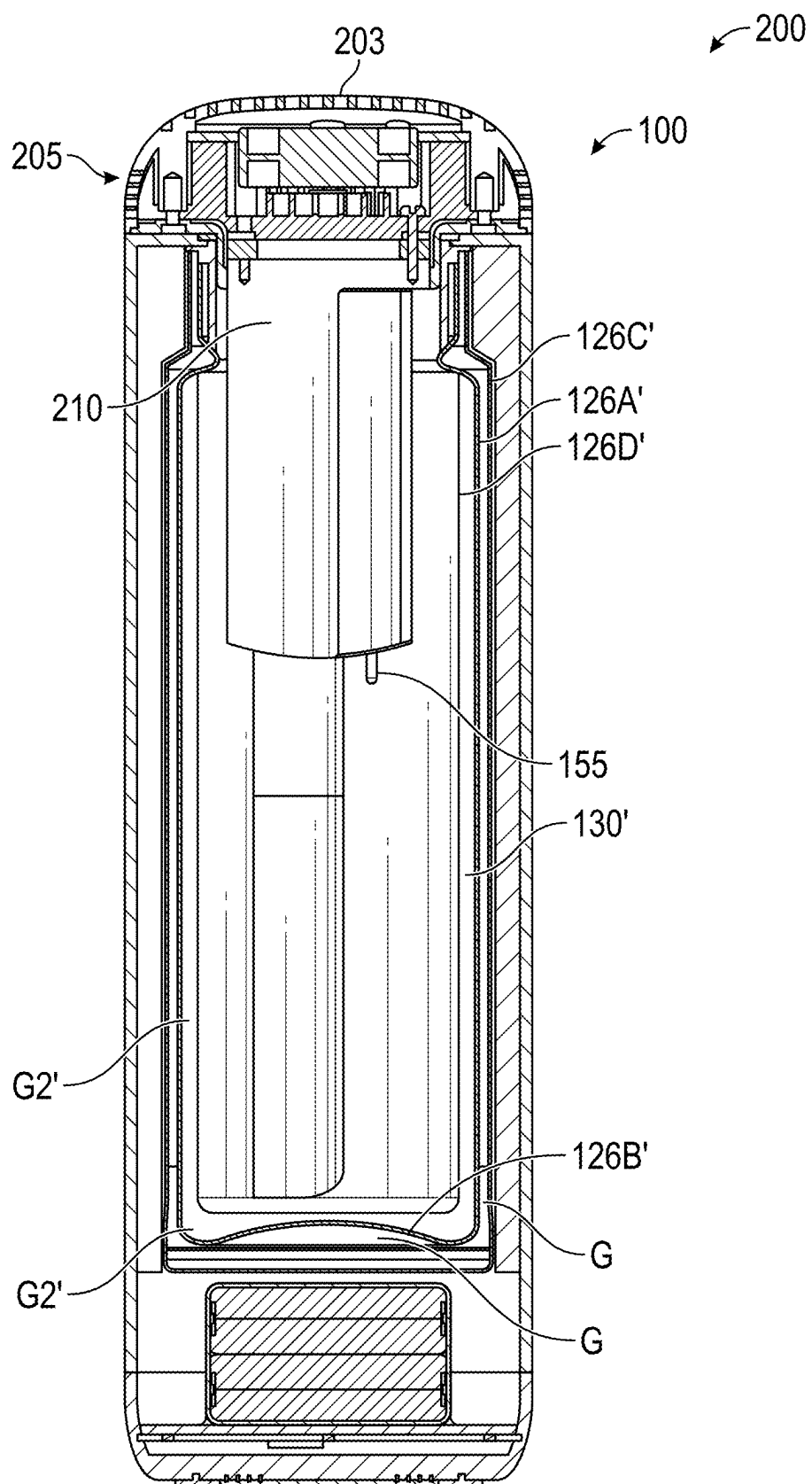


FIG. 7

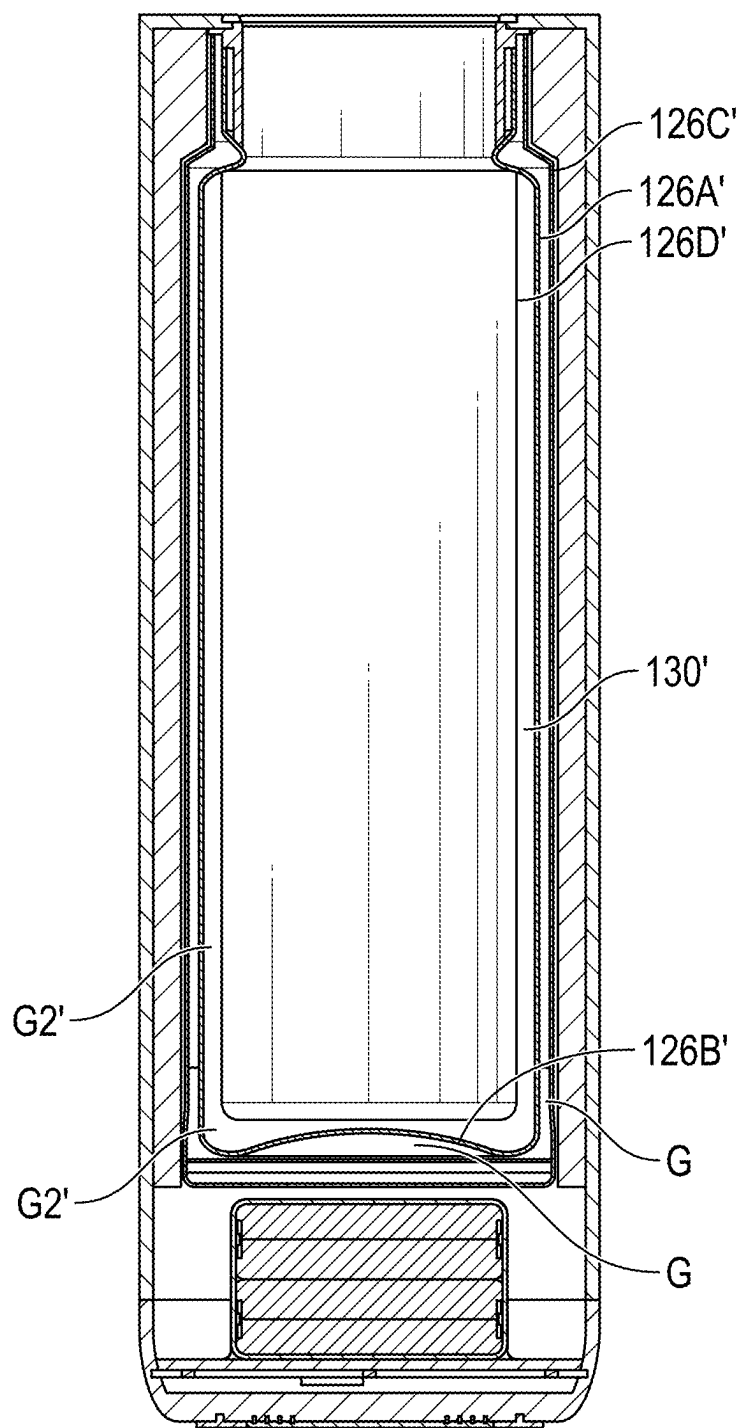


FIG. 8

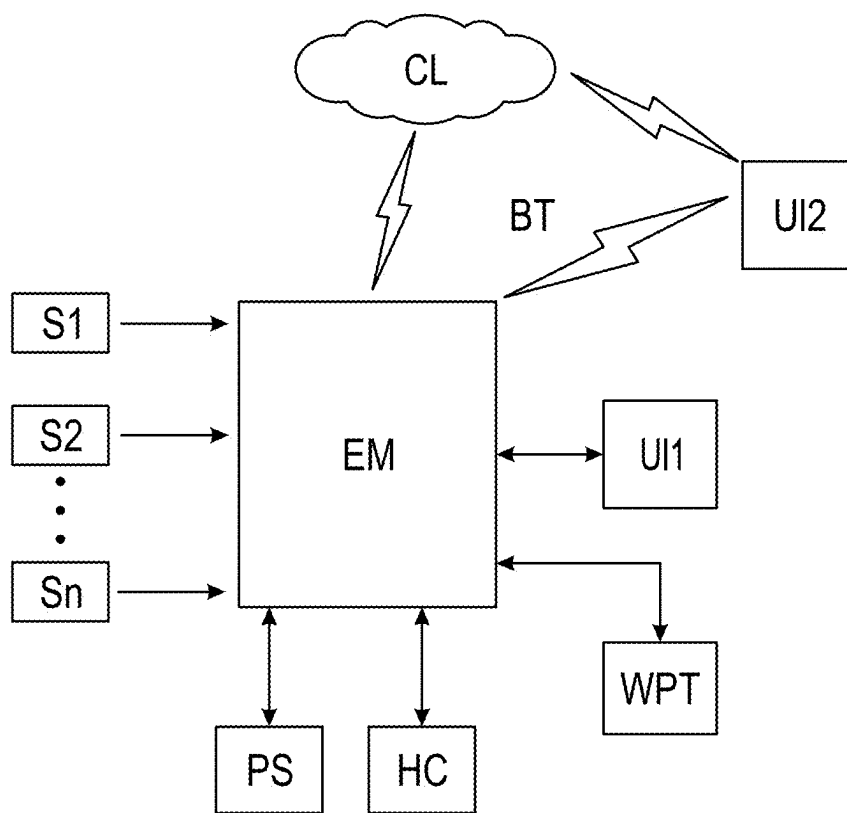


FIG. 9

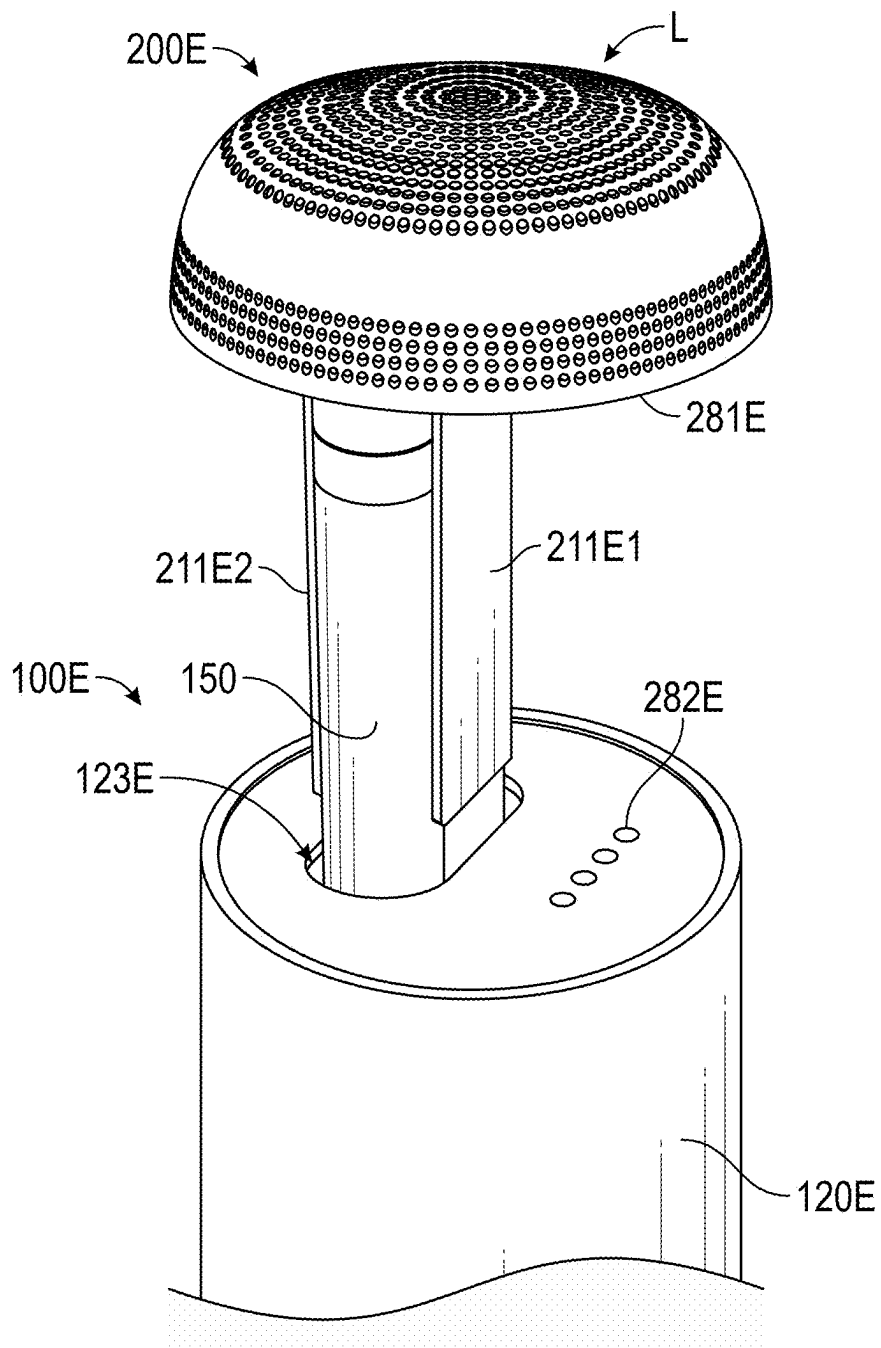


FIG. 10A

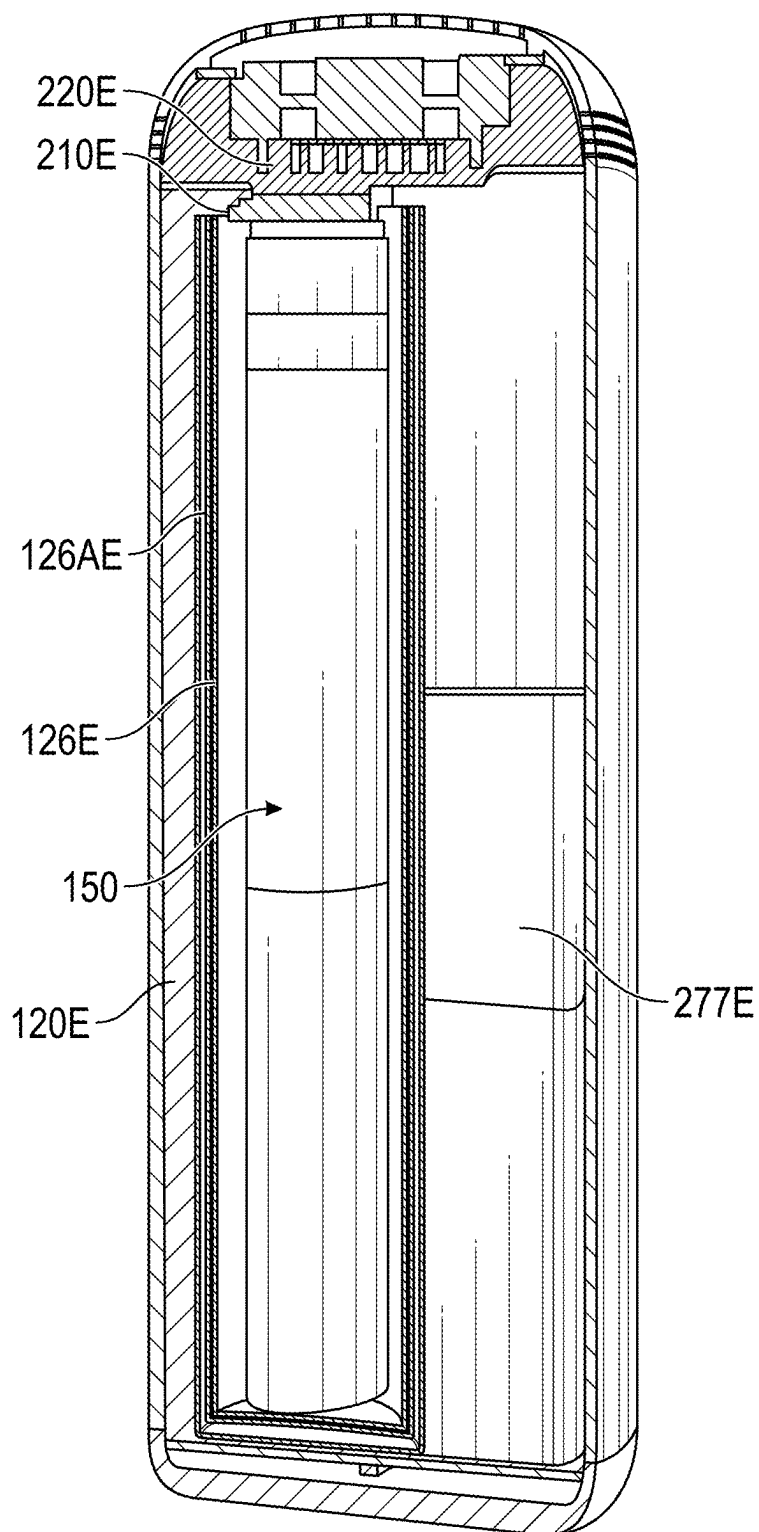


FIG. 10B

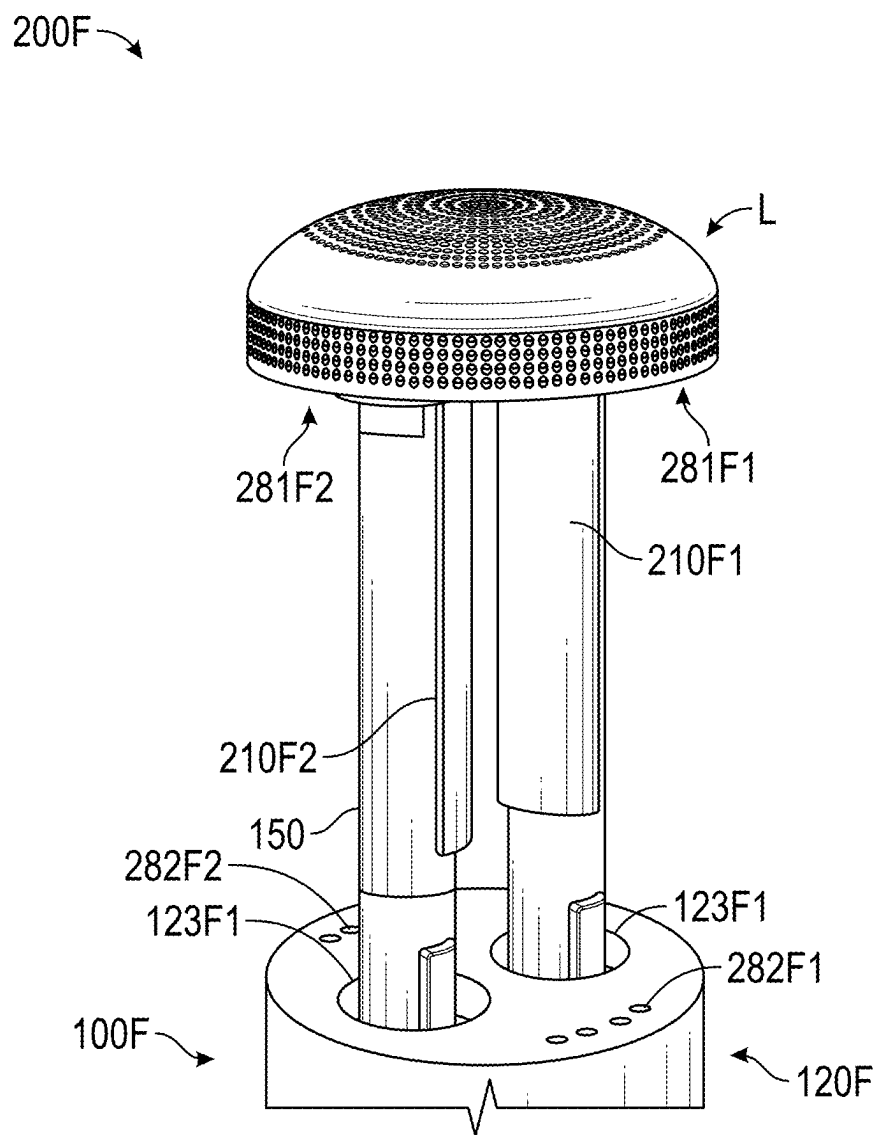


FIG. 11A

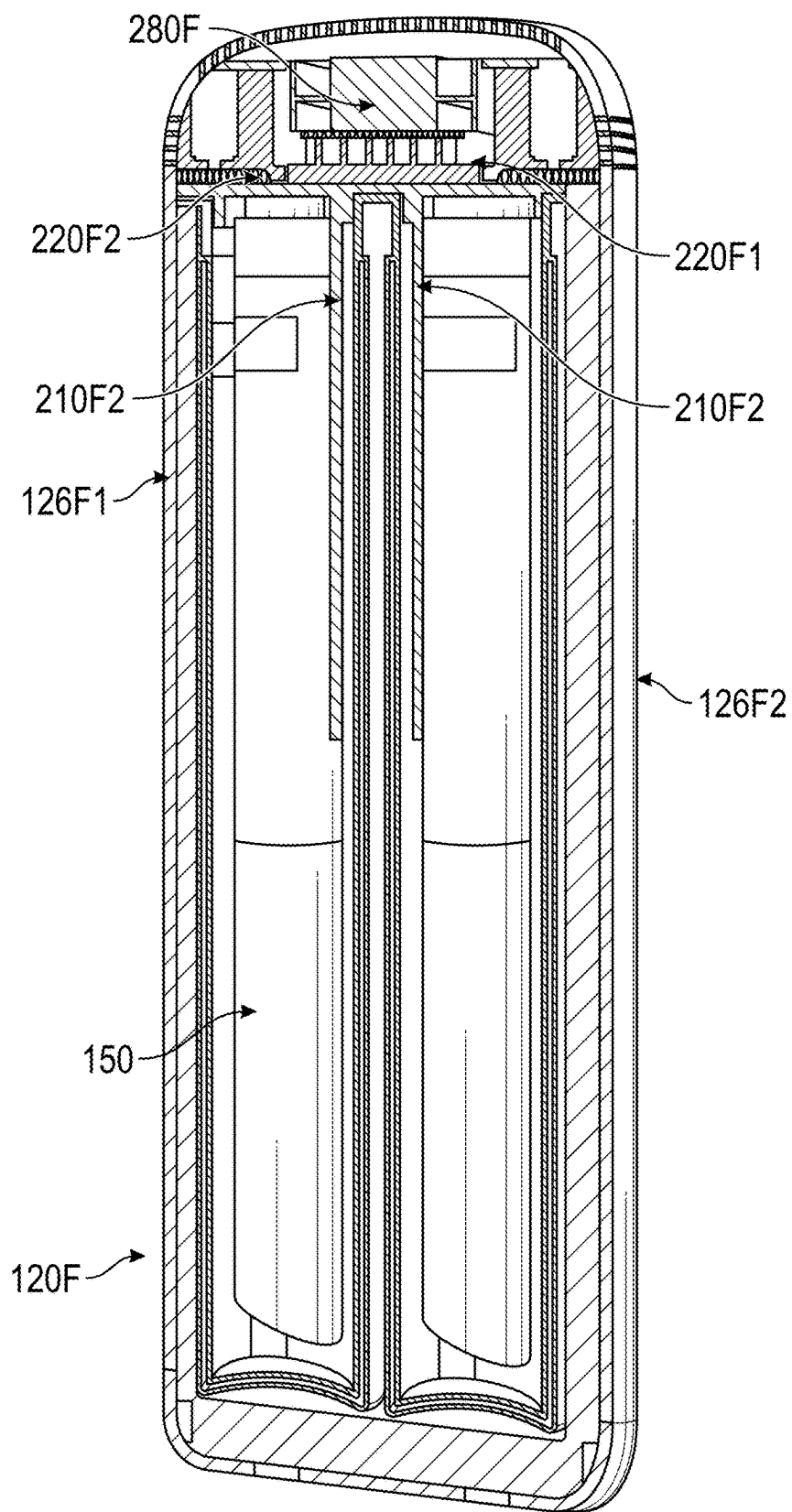


FIG. 11B

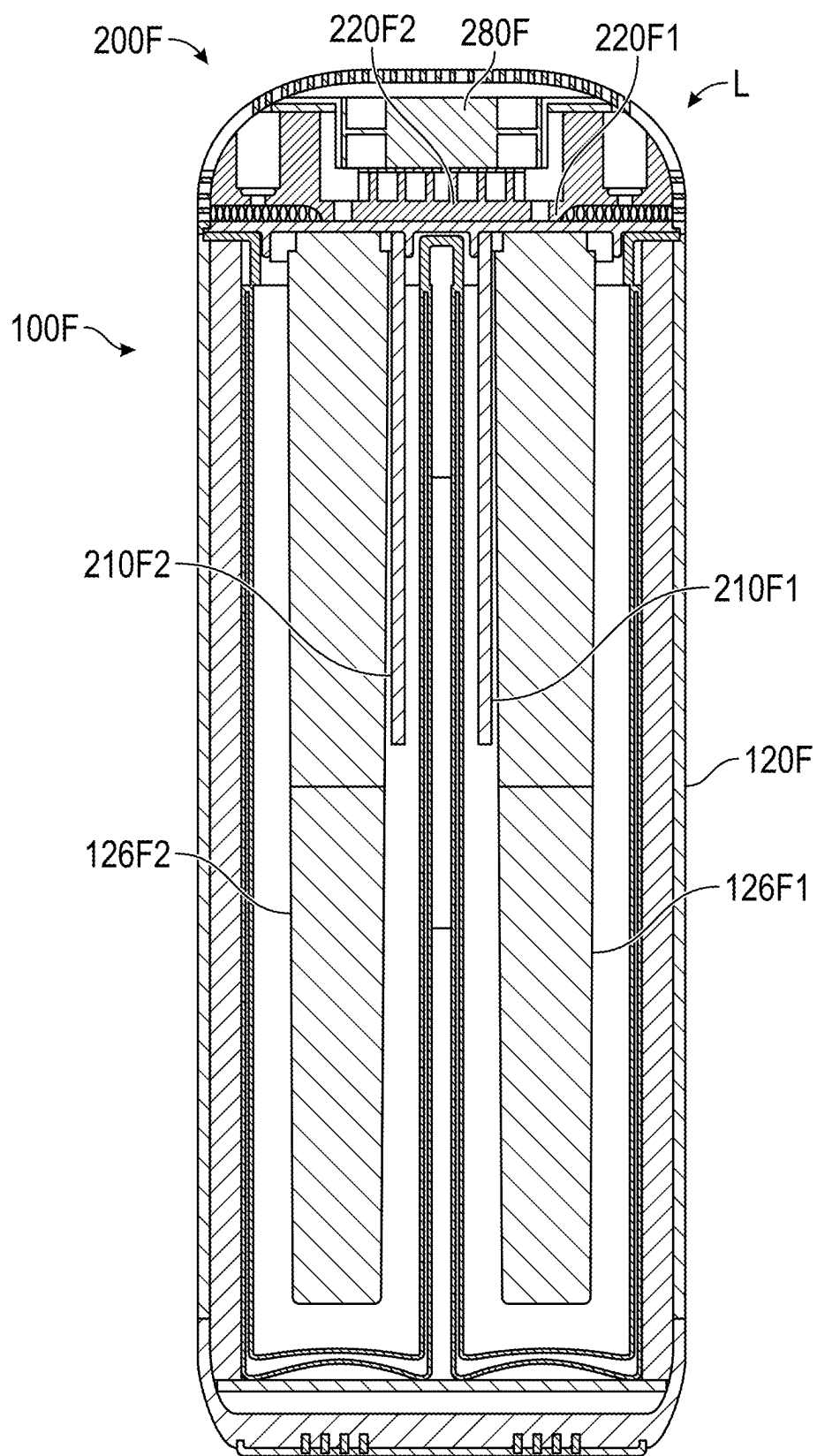


FIG. 11C

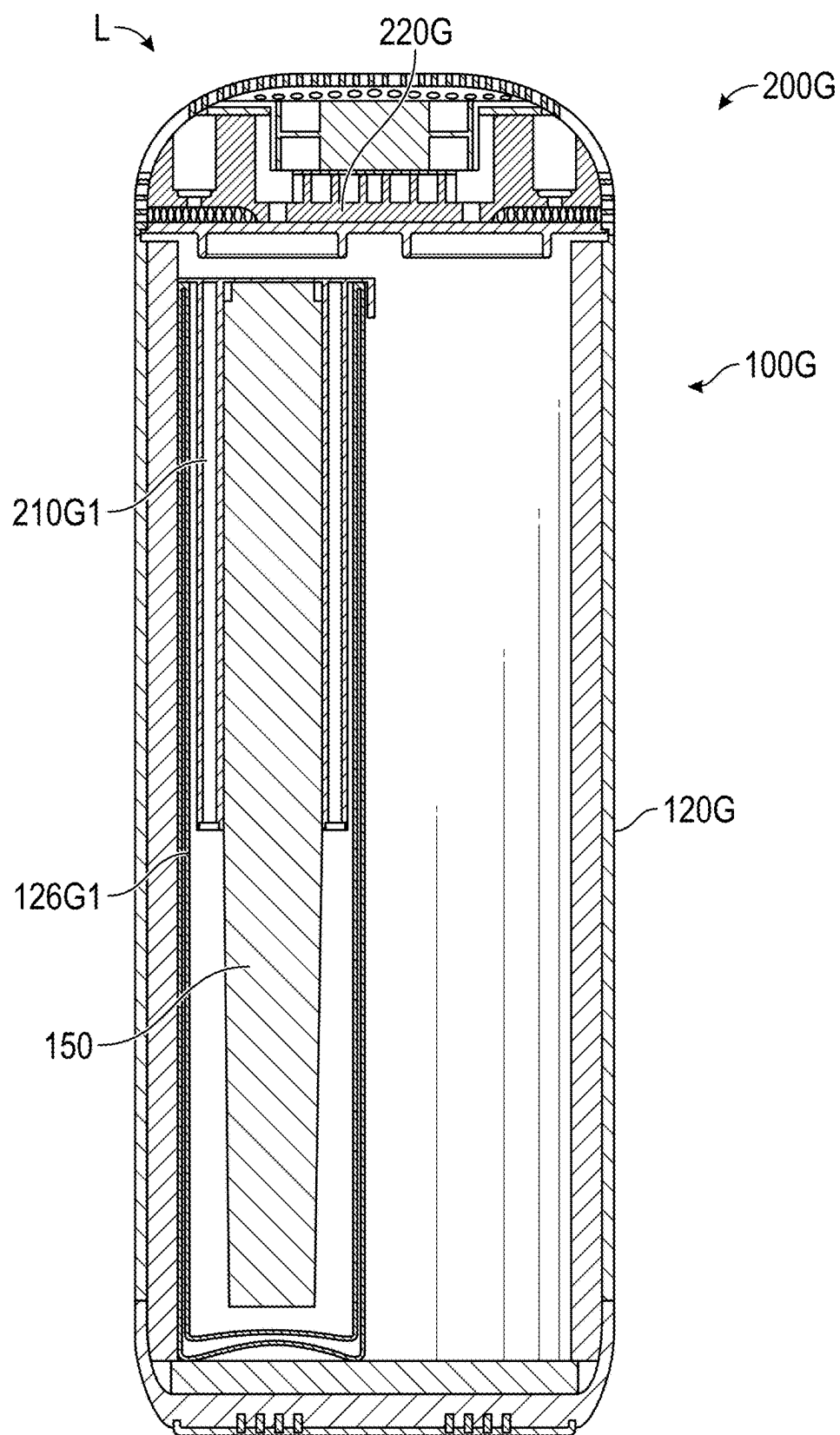


FIG. 12A

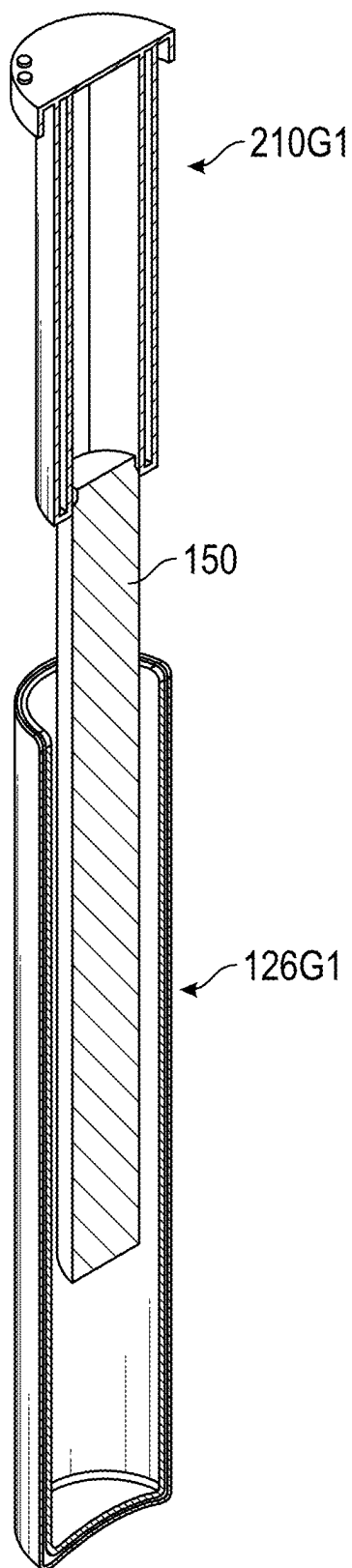


FIG. 12B

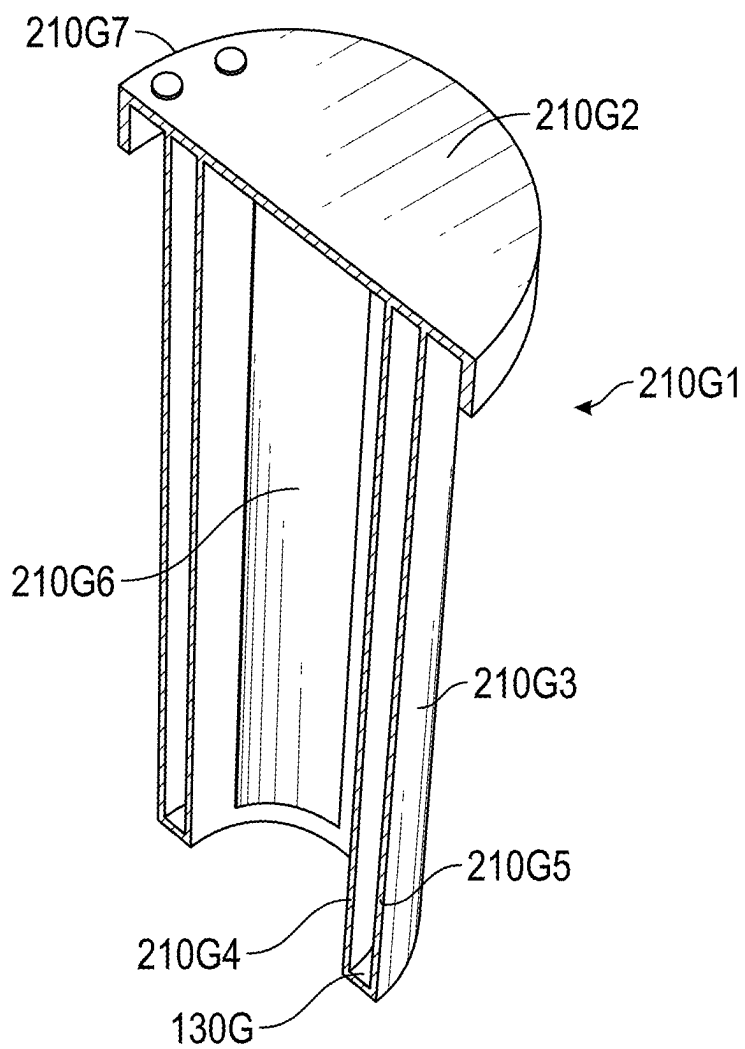


FIG. 12C

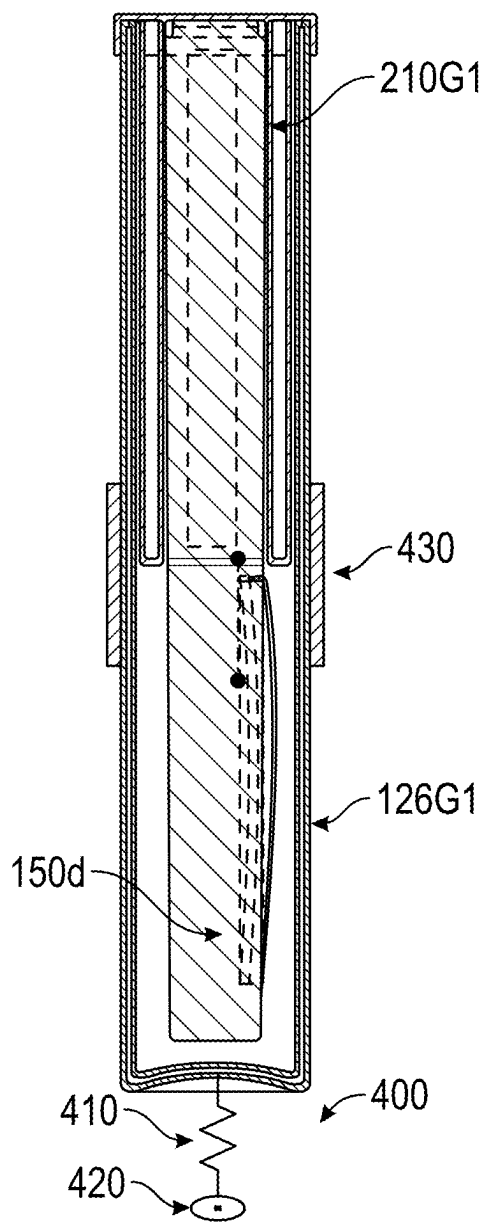


FIG. 13

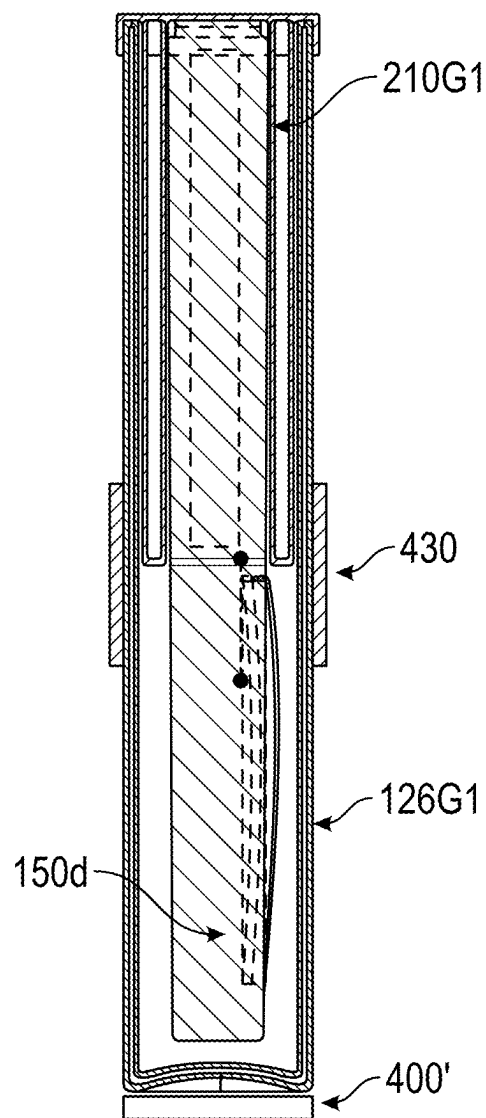


FIG. 14A

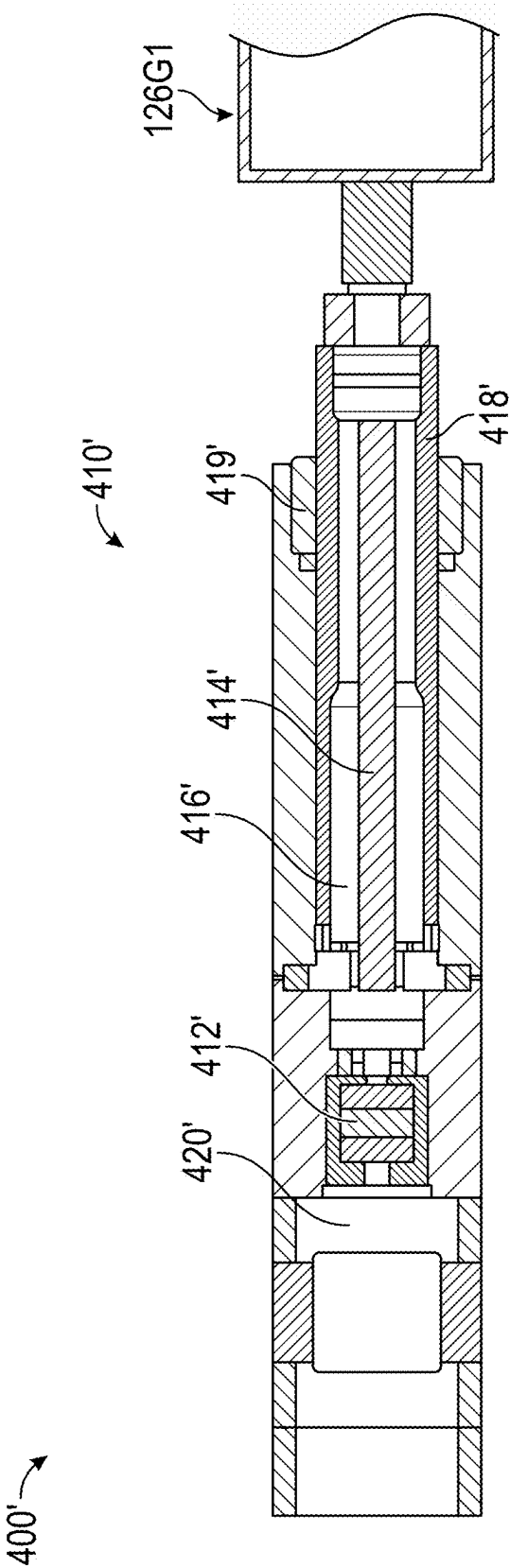


FIG. 14B

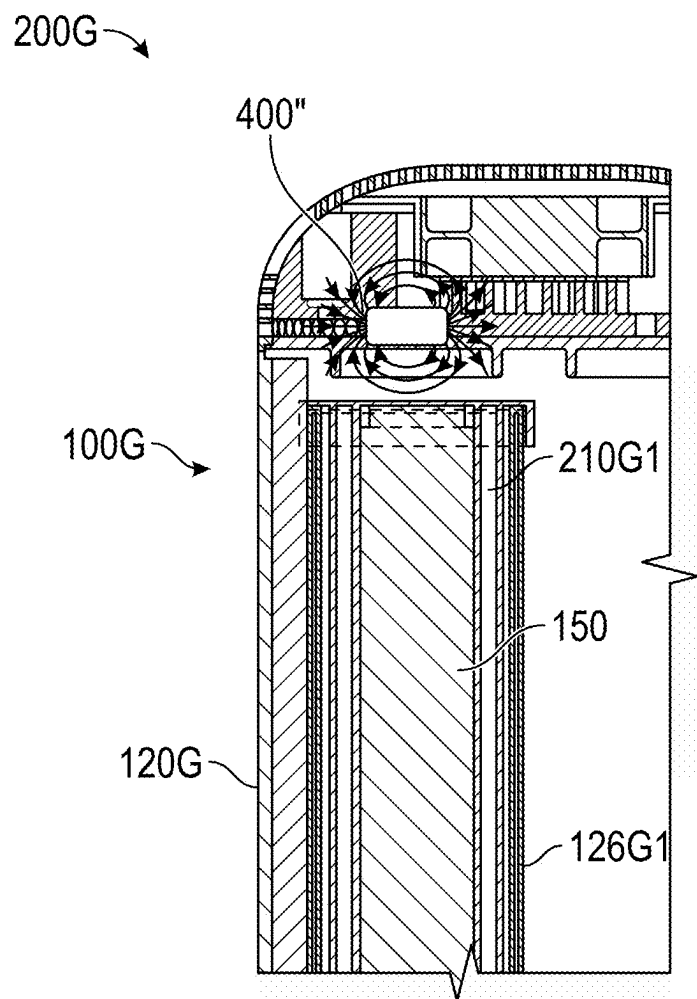


FIG. 15

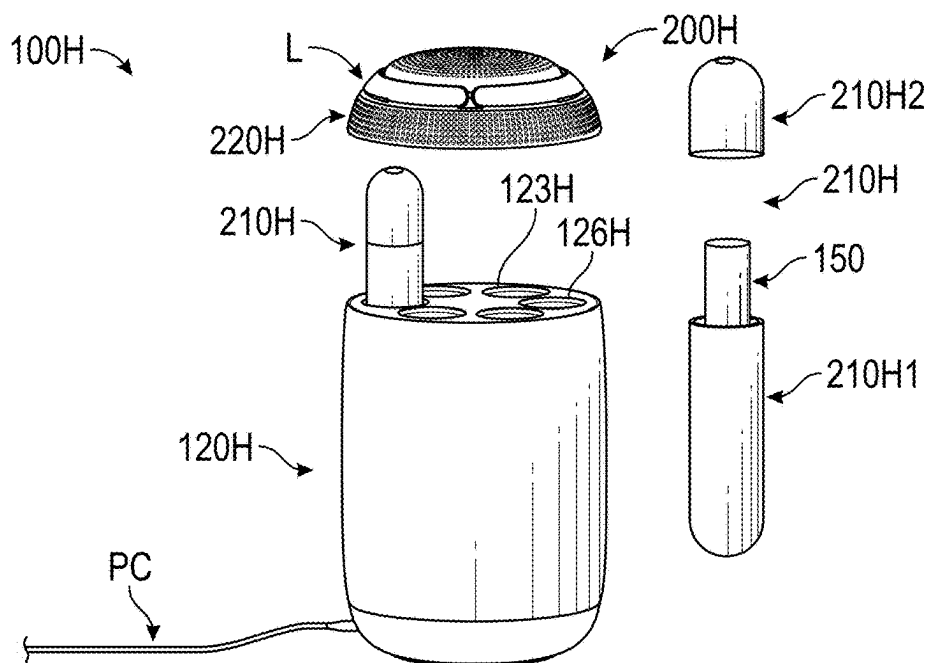


FIG. 16

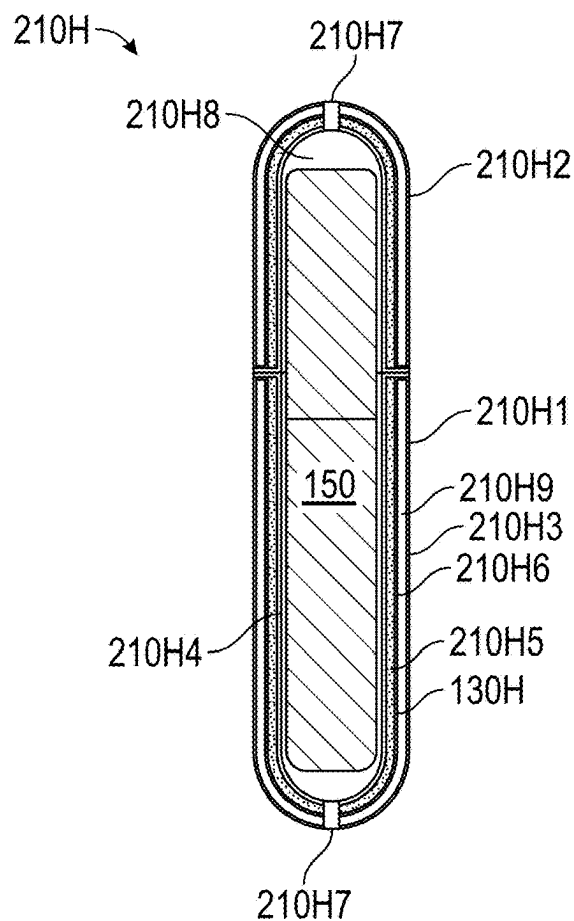


FIG. 16A

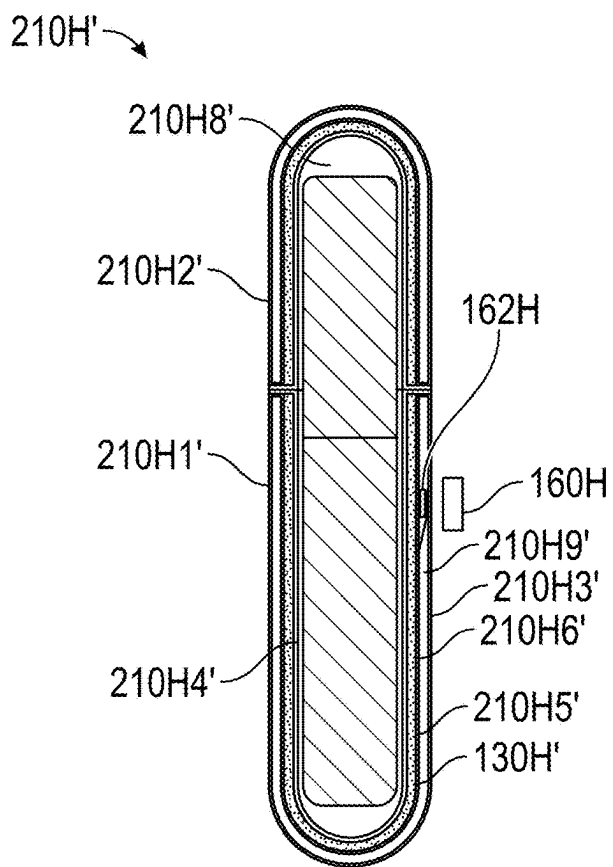


FIG. 16B

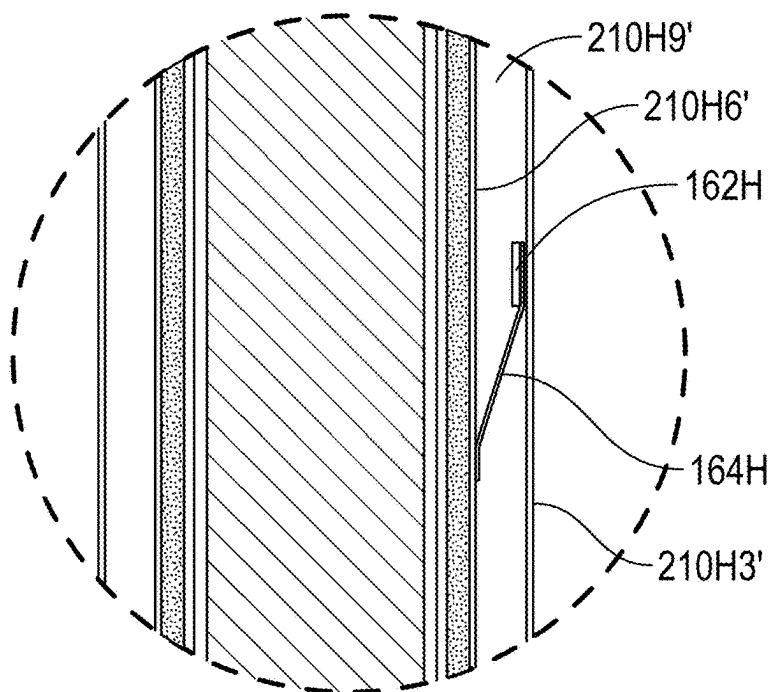


FIG. 16C

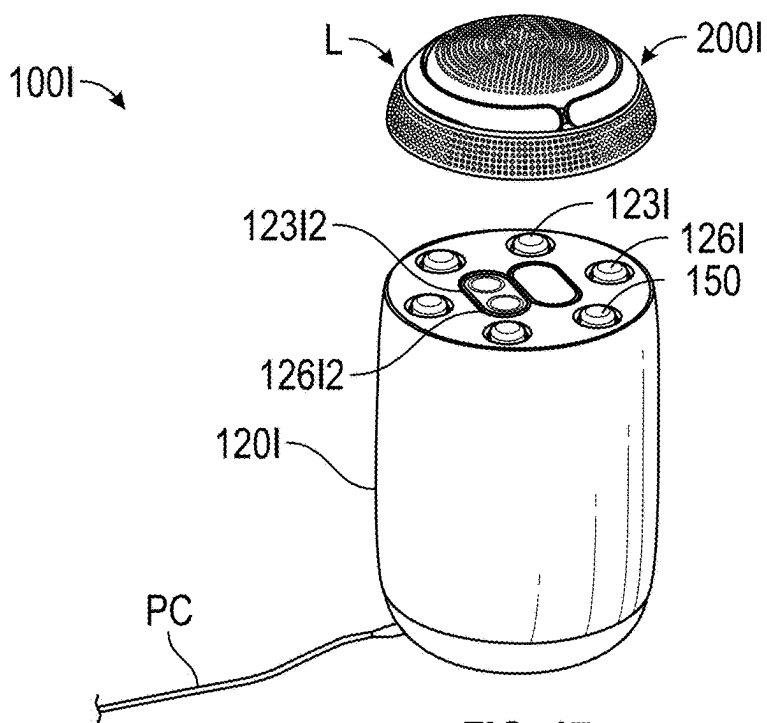


FIG. 17

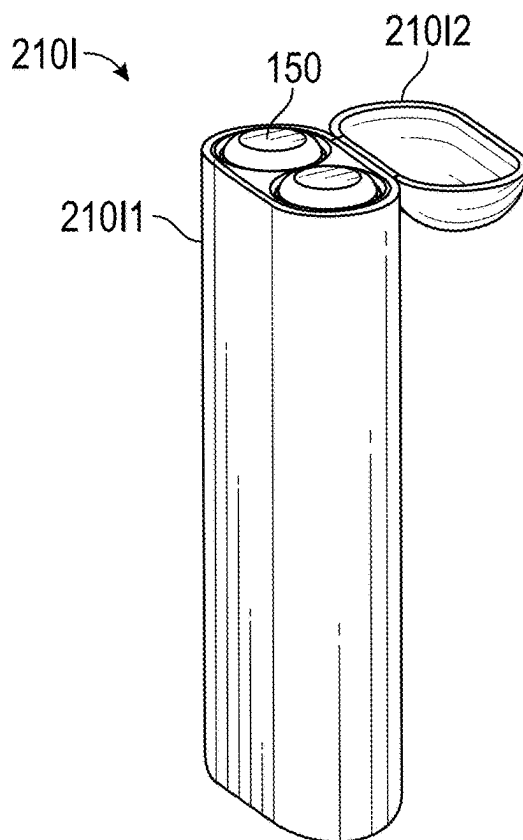


FIG. 17A

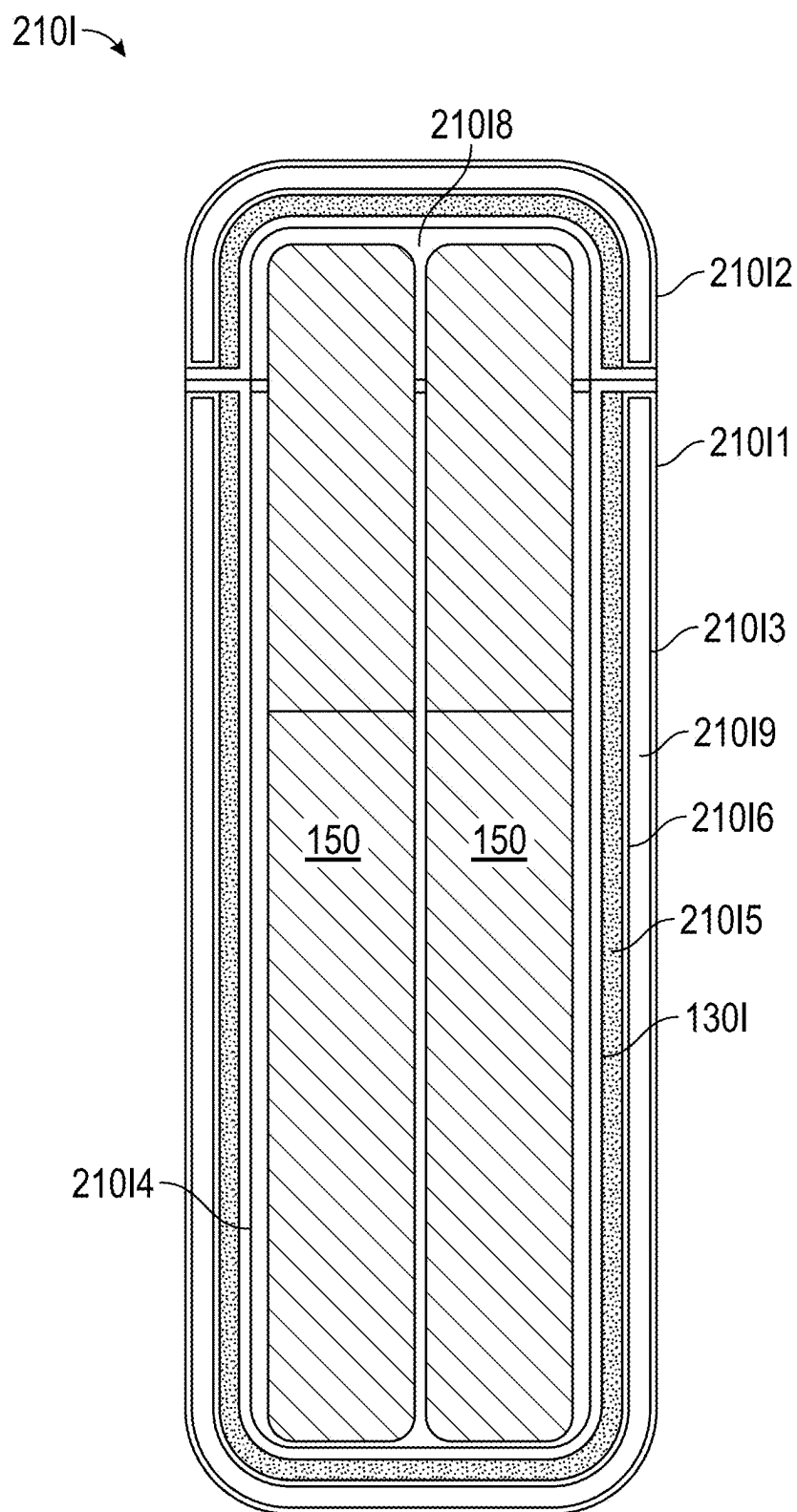


FIG. 17B

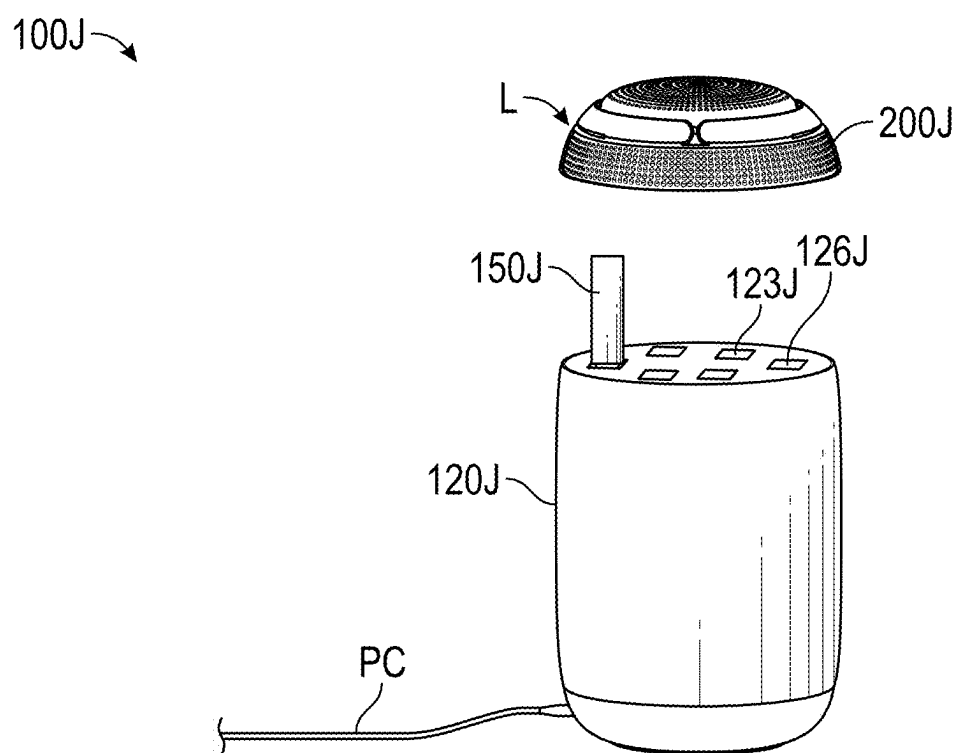


FIG. 18

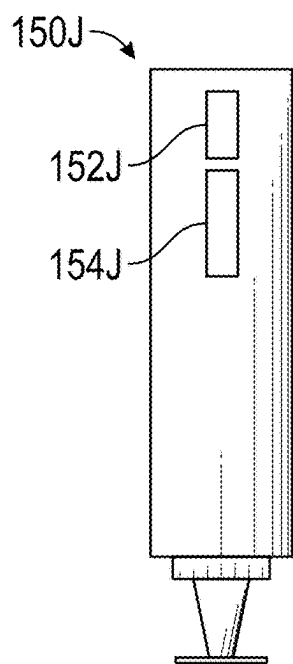


FIG. 18A

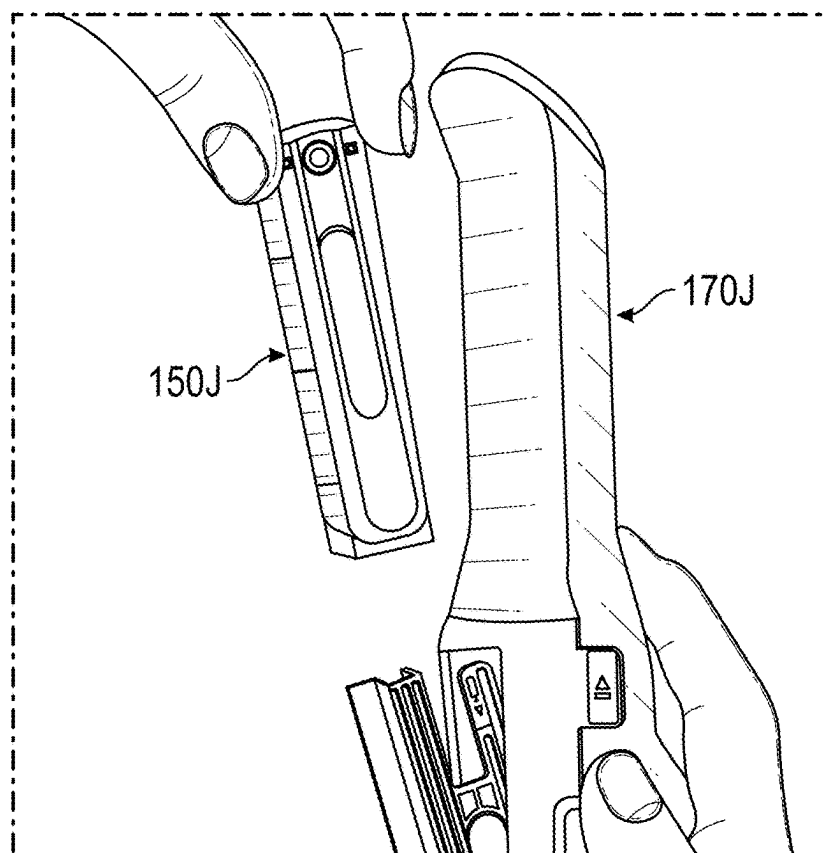


FIG. 18B

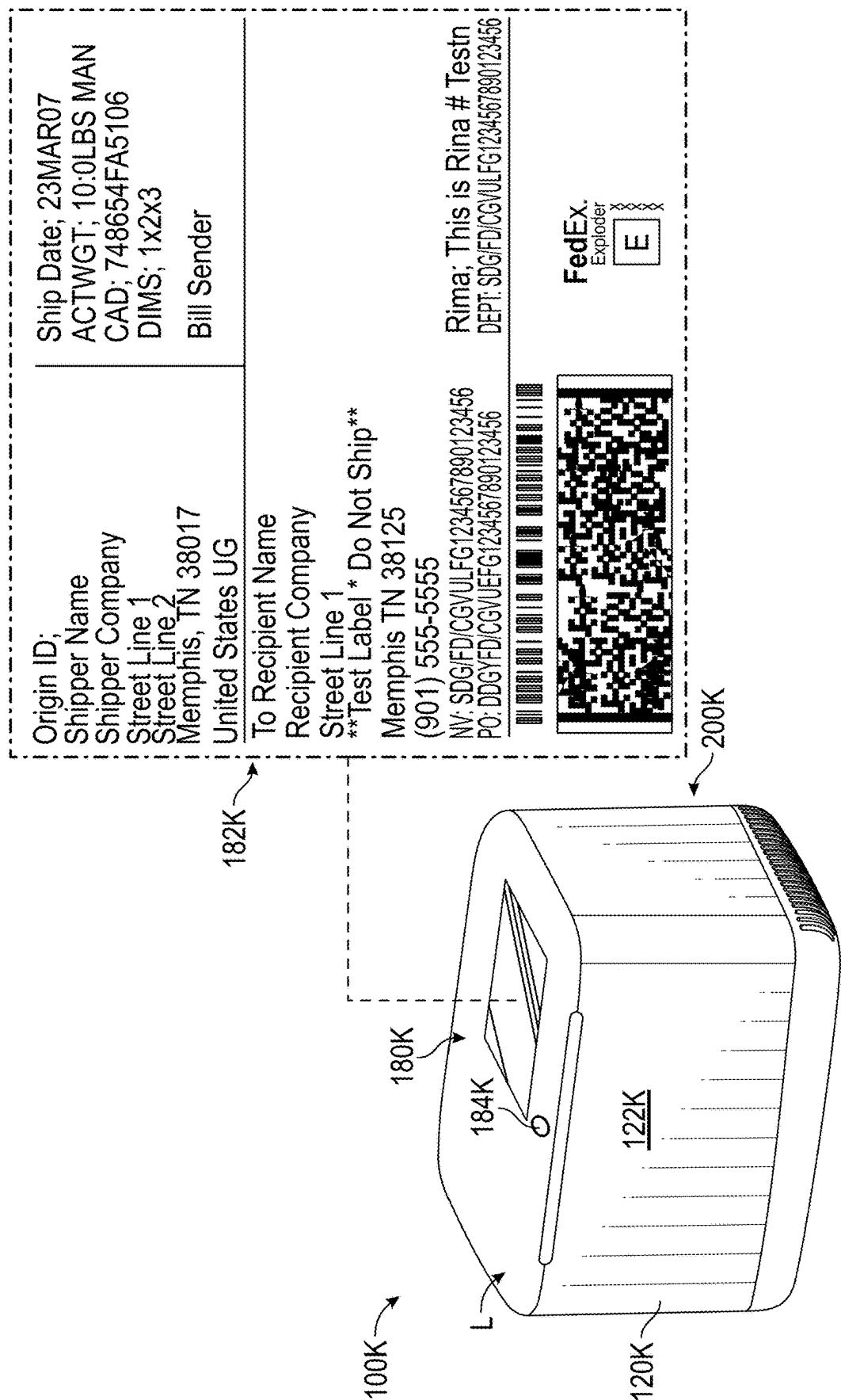


FIG. 19A

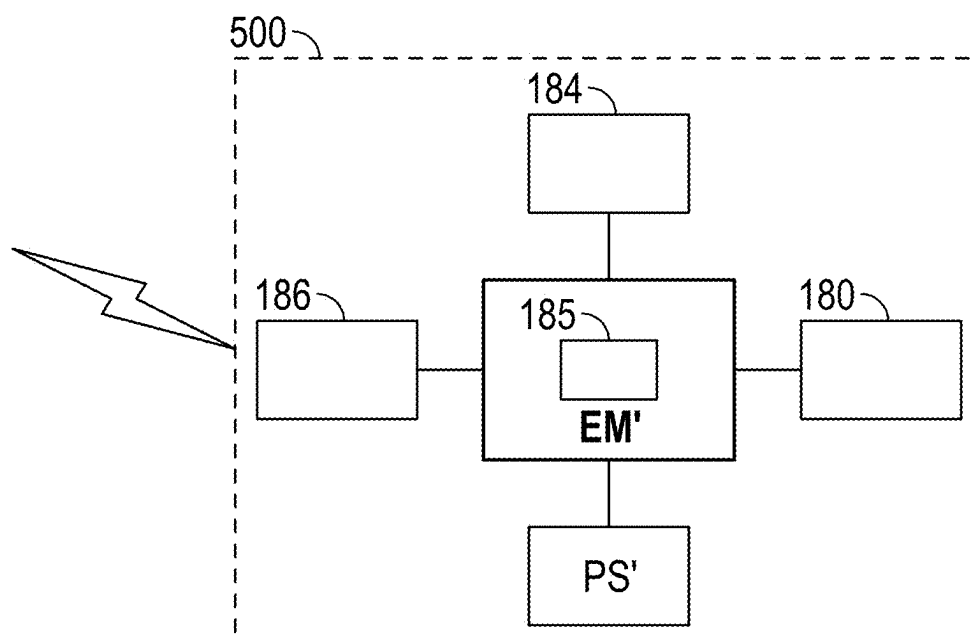


FIG. 19B

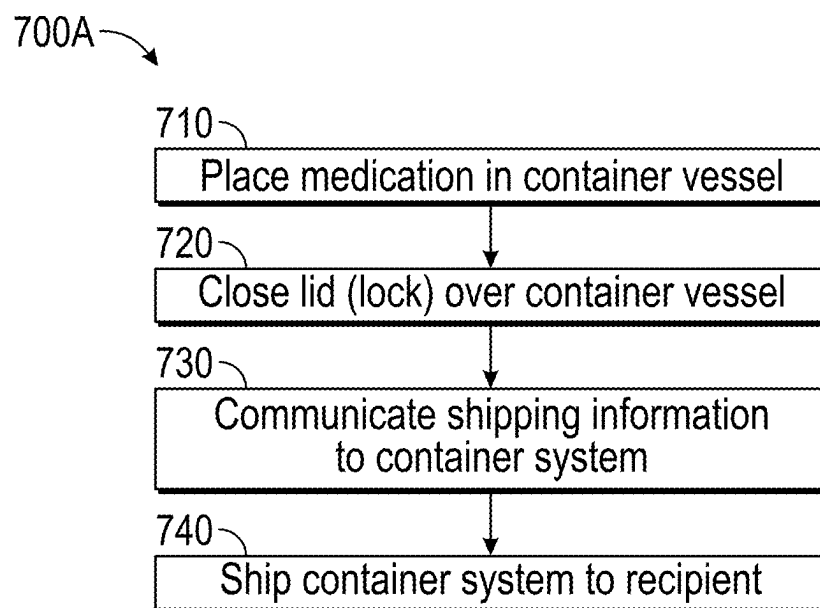


FIG. 20A

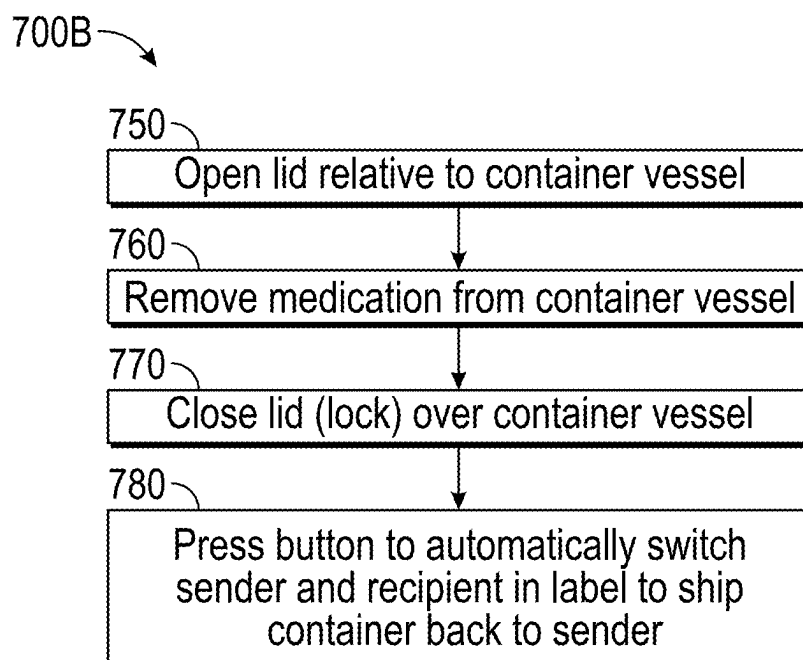


FIG. 20B

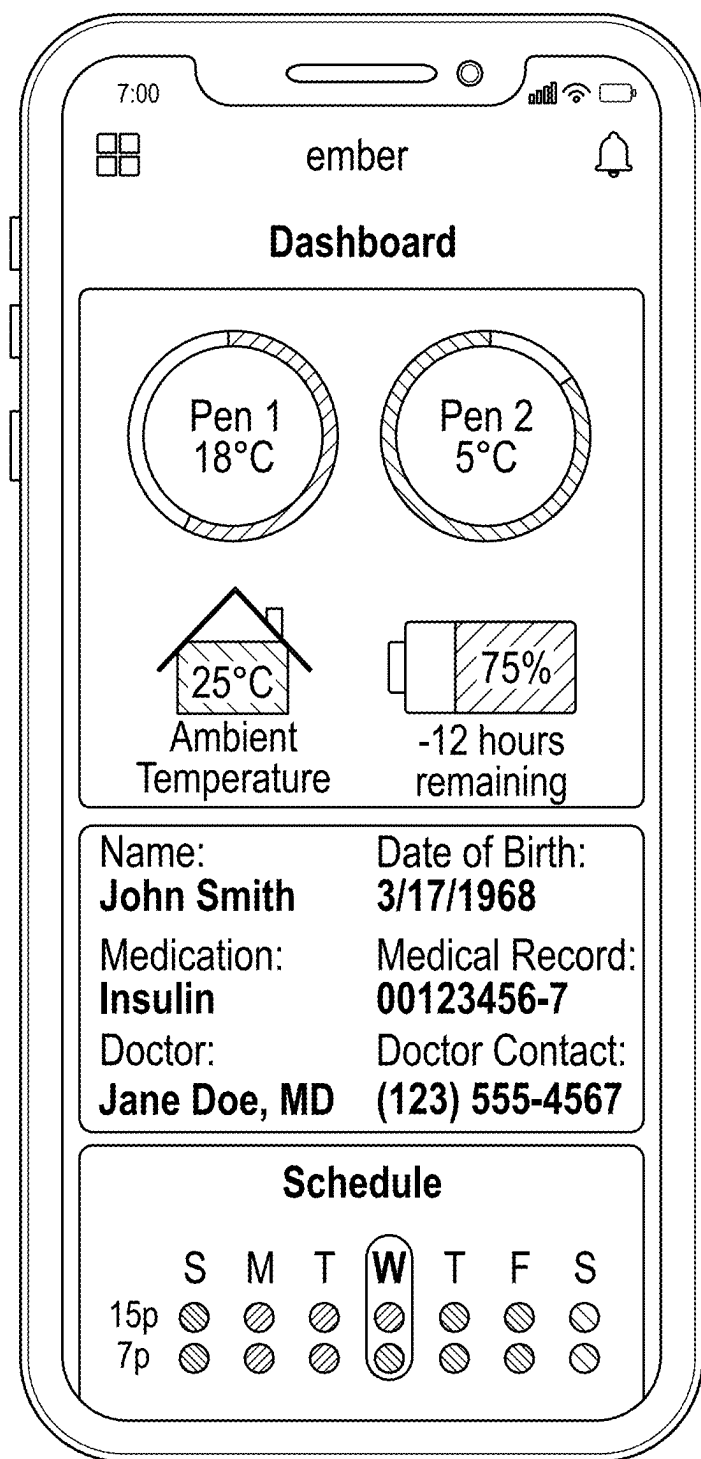


FIG. 21A

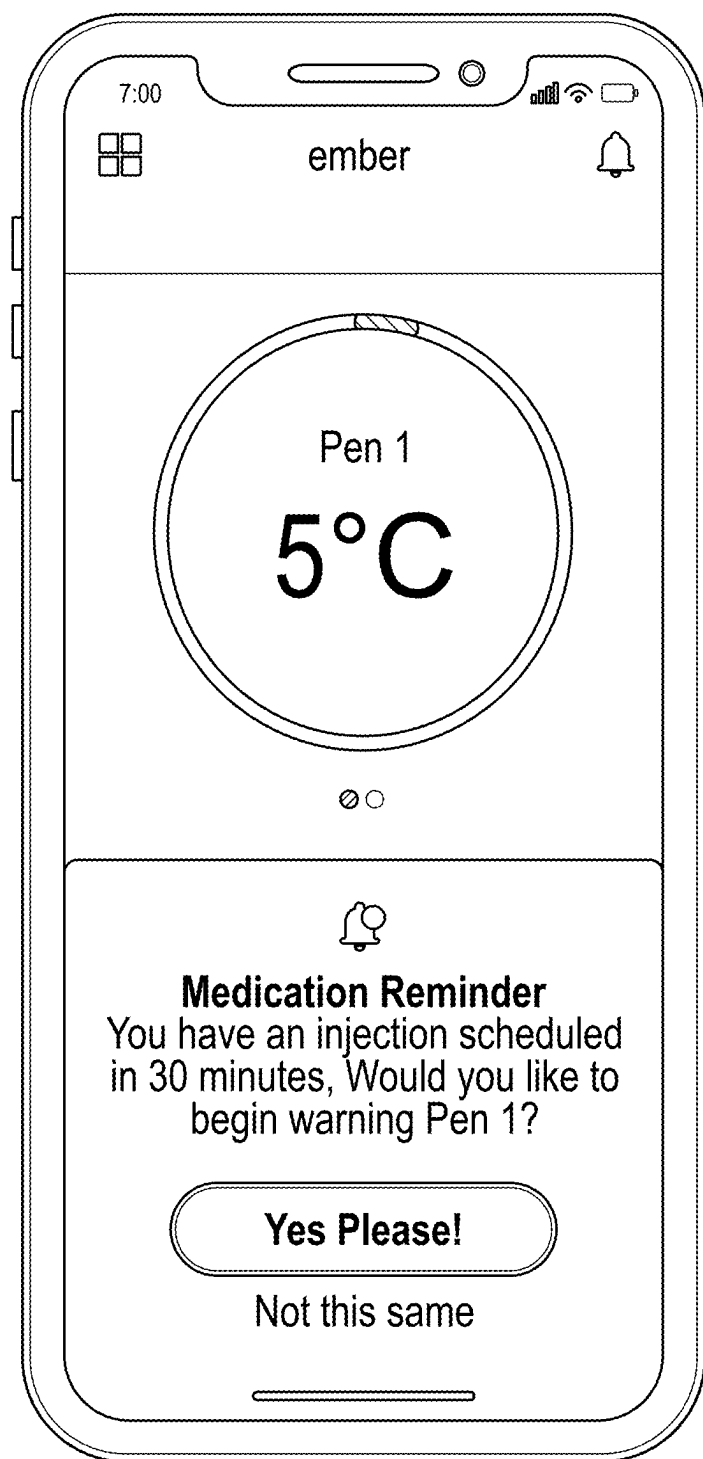


FIG. 21B

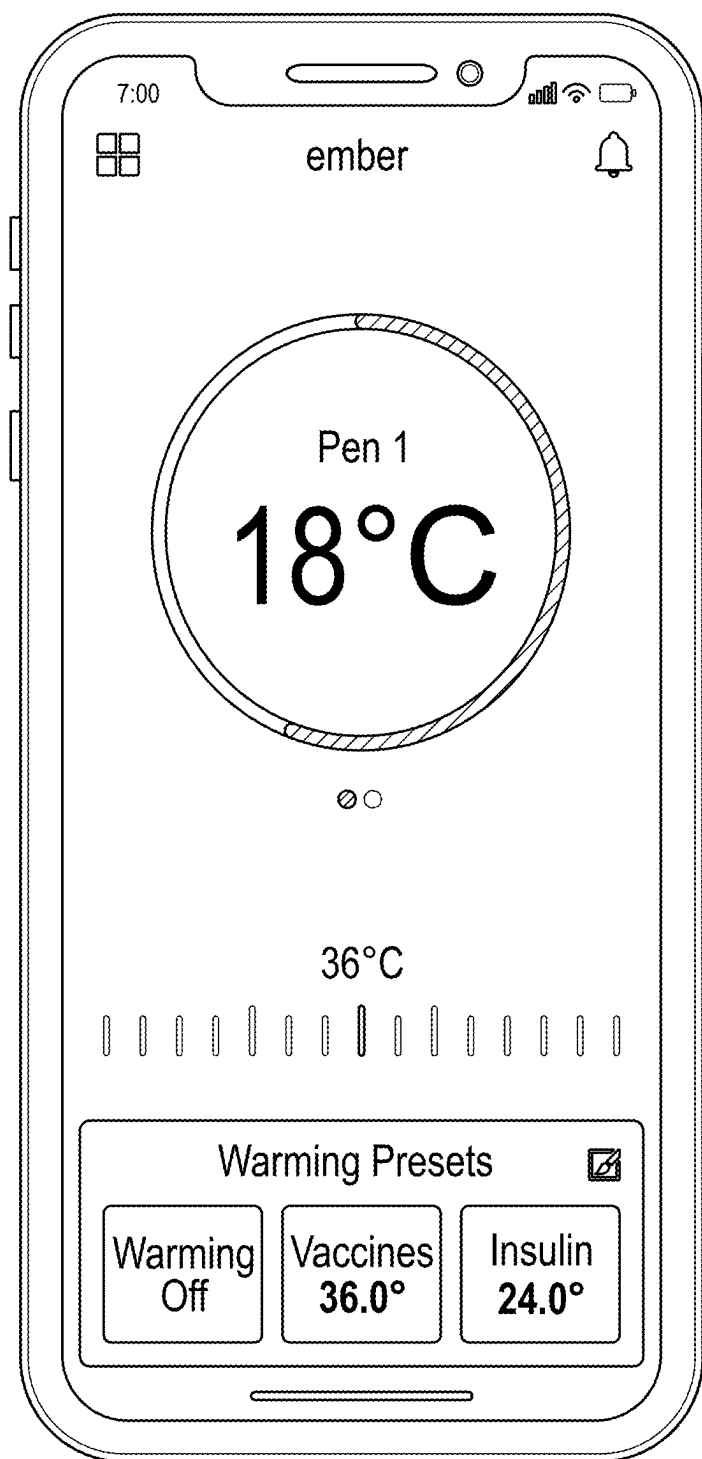


FIG. 21C

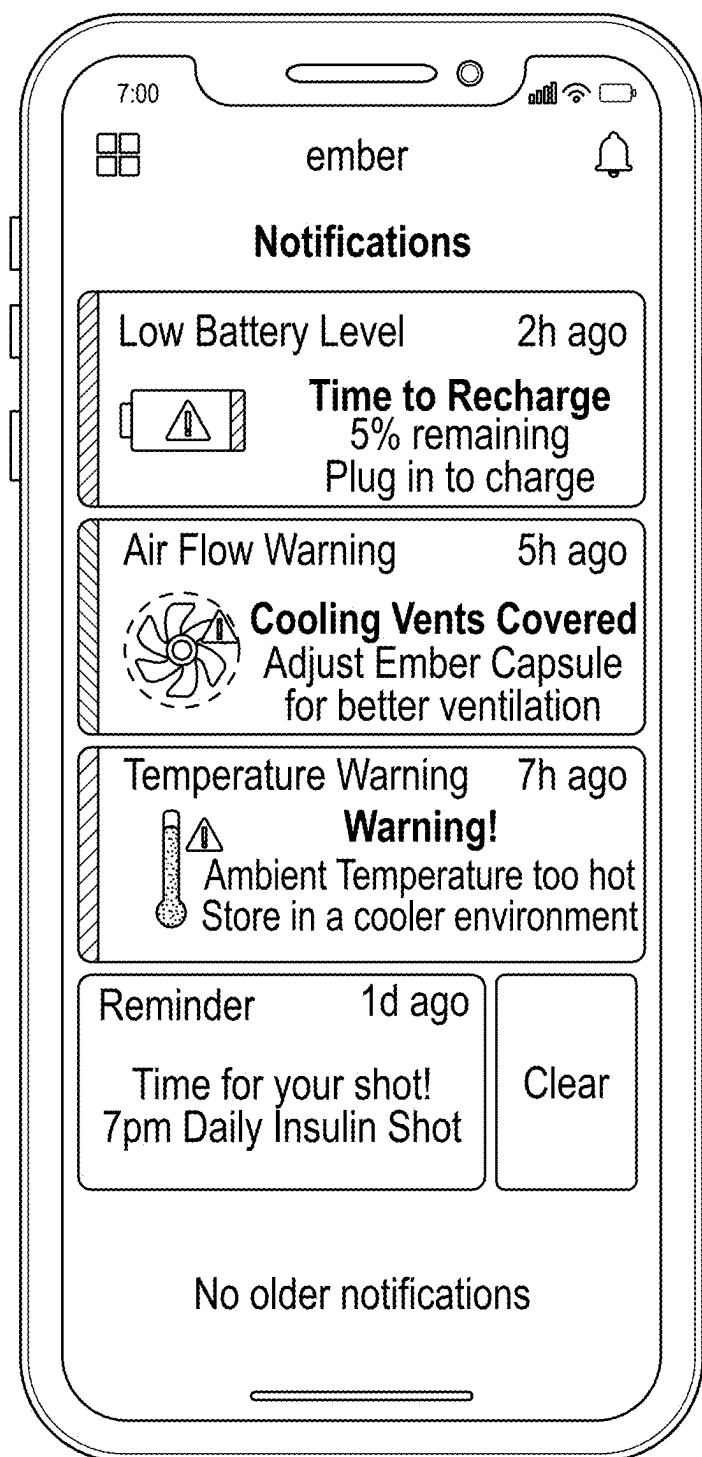


FIG. 21D

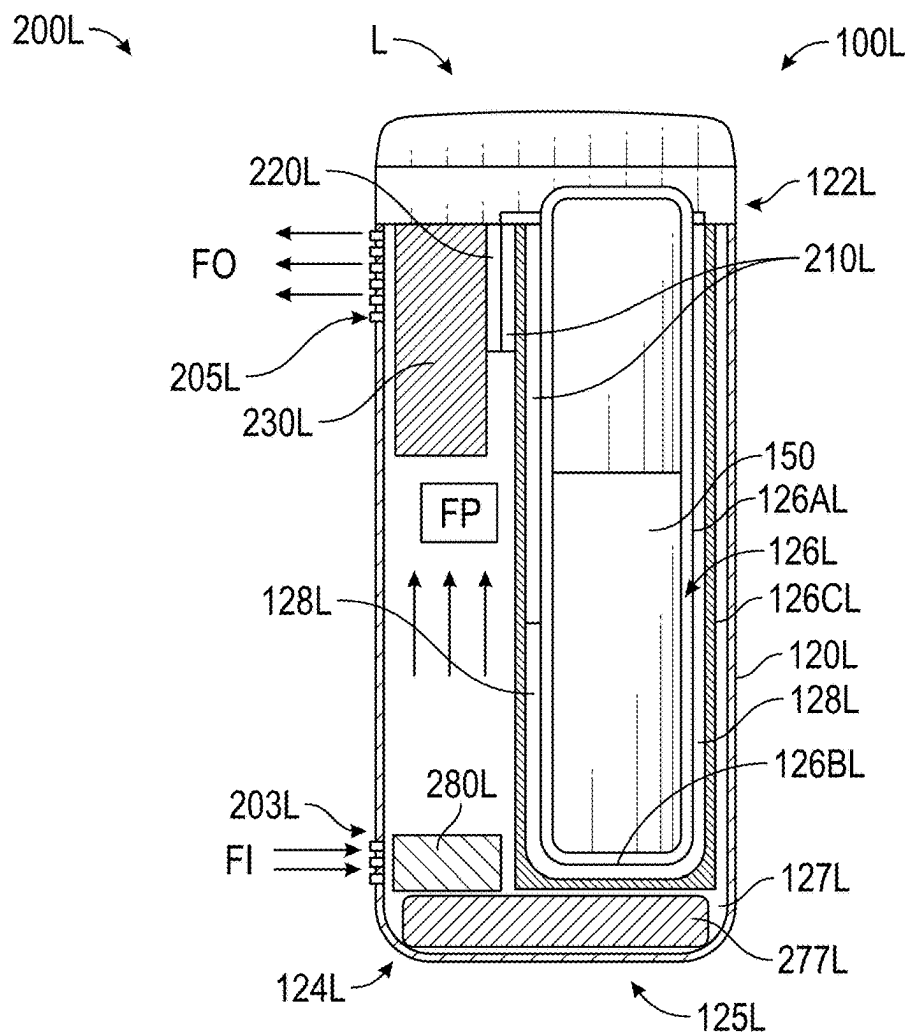


FIG. 22A

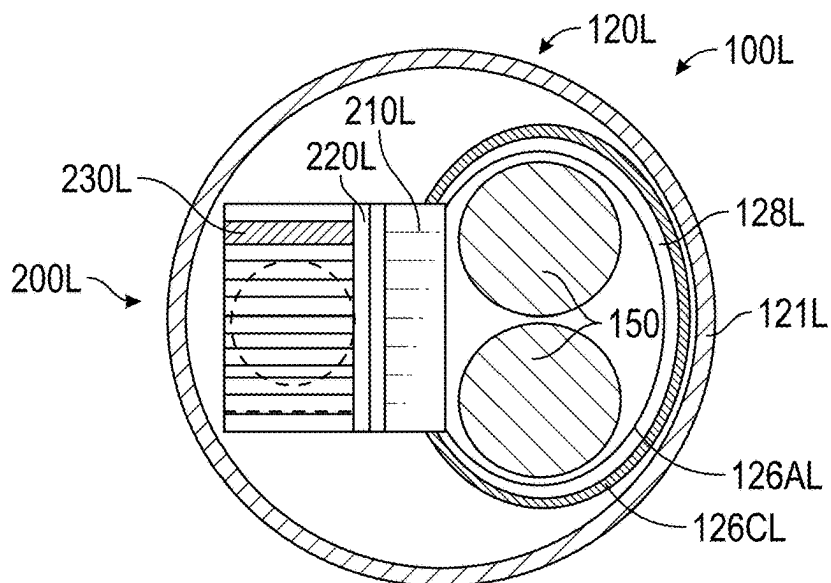


FIG. 22B

PORTABLE COOLER WITH ACTIVE TEMPERATURE CONTROL

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

[0001] Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57 and should be considered a part of this specification.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The invention is directed to a portable cooler (e.g., for medicine such as insulin, vaccines, epinephrine, etc.), and more particularly to a portable cooler with active temperature control.

Description of the Related Art

[0003] Certain medicine needs to be maintained at a certain temperature or temperature range to be effective (e.g., to maintain potency). Once potency of medicine (e.g., a vaccine, insulin, epinephrine) is lost, it cannot be restored, rendering the medicine ineffective and/or unusable. For example, injector pens are commonly used to deliver medication, such as epinephrine to counteract the effects of an allergic reaction (e.g., due to a peanut allergy, insect stings/bites, etc.). Users sometimes carry such medicine (e.g., medicine injector pens, cartridges for injector pens) with them (e.g., in a bag, purse, pocket, etc.) in the event they suffer an allergic reaction during the day. However, such medicine may be exposed to varying temperatures during the day (e.g., due to ambient temperature conditions, temperature conditions in the car, workplace, school, etc.), which can be outside the preferred temperature or temperature range for the medicine to be effective.

SUMMARY

[0004] Accordingly, there is a need for improved portable cooler designs (e.g., for storing and/or transporting medicine, such as epinephrine, vaccines, insulin, etc.) that can maintain the contents of the cooler at a desired temperature or temperature range. Additionally, there is a need for an improved portable cooler design with improved cold chain control and record keeping of the temperature history of the contents (e.g., medicine, such as epinephrine, vaccines, insulin, etc.) of the cooler (e.g., during storage and/or transport of the medicine, such as during a commute to work or school).

[0005] In accordance with one aspect, a portable cooler container (e.g., capsule) with active temperature control system is provided. The active temperature control system is operated to heat or cool a chamber of a vessel to approach a temperature set point suitable for a medication (e.g., epinephrine, insulin, vaccines, etc.) stored in the cooler container.

[0006] In accordance with another aspect, a portable cooler (or capsule) is provided that includes a temperature control system operable (e.g., automatically operable) to maintain the chamber of the cooler at a desired temperature or temperature range for a prolonged period of time. Optionally, the portable cooler is sized to house one or more

containers (e.g., injector pens and/or cartridges for injector pens, vials, etc.). Optionally, the portable cooler automatically logs (e.g., stores on a memory of the cooler) and/or communicates data on one or more sensed parameters (e.g., of the temperature of the chamber, battery charge level, etc.) to a remote electronic device (e.g., remote computer, mobile electronic device such as a smartphone or tablet computer). Optionally, the portable cooler can automatically log and/or transmit the data to the remote electronic device (e.g., automatically in real time, periodically at set intervals, etc.).

[0007] In accordance with another aspect, a portable cooler container (e.g., capsule) with active temperature control is provided. The container comprises a container body having a chamber configured to receive and hold one or more containers (e.g., injector pens, cartridges for injector pens, vials, etc.), the chamber defined by a base and an inner peripheral wall of the container body. The container also comprises a temperature control system comprising one or more thermoelectric elements (e.g., Peltier elements) configured to actively heat or cool a heat sink component in thermal communication (e.g., in contact with) the one or more containers (e.g., medicine containers) in the chamber, and circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the heat sink component and/or chamber to a predetermined temperature or temperature range.

[0008] Optionally, the container can include one or more batteries configured to provide power to one or both of the circuitry and the one or more thermoelectric elements.

[0009] Optionally, the circuitry is further configured to wirelessly communicate with a cloud-based data storage system (e.g., remote server) or a remote electronic device (e.g., smartphone, tablet computer, laptop computer, desktop computer).

[0010] Optionally, the container includes a first heat sink in thermal communication with the chamber, the first sink being selectively thermally coupled to the one or more thermoelectric elements. Optionally, the first heat sink can removably extend into the chamber of the container and one or more containers (e.g., medicine containers, such as injector pens, cartridges for injector pens, vials, etc.) can releasably couple to the first heat sink (e.g., to one or more clip portions or slots of the first heat sink) so that the one or more containers are disposed in the chamber.

[0011] Optionally, the container includes a second heat sink in communication with the one or more thermoelectric elements (TECs), such that the one or more TECs are disposed between the first heat sink and the second heat sink.

[0012] Optionally, the second heat sink is in thermal communication with a fan operable to draw heat from the second heat sink.

[0013] In one implementation, such as where the ambient temperature is above the predetermined temperature or temperature range, the temperature control system is operable to draw heat from the first heat sink (and draw heat from the chamber), which transfers said heat to the one or more TECs, which transfer said heat to the second heat sink, where the optional fan dissipates heat from the second heat sink. The temperature control system can in this manner cool the first heat sink (and the chamber), thereby cooling the containers (e.g., medicine containers) in the chamber toward the predetermined temperature or temperature range.

[0014] In another implementation, such as where the ambient temperature is below the predetermined temperature

ture or temperature range, the temperature control system is operable to add heat to the first heat sink (and add heat to the chamber), which transfers said heat from the one or more TECs. The temperature control system can in this matter heat the first heat sink (and the chamber), thereby heating the containers (e.g., medicine containers) in the chamber toward the predetermined temperature or temperature range.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. 1 is a schematic view of one embodiment of a cooler container.

[0016] FIG. 2 is a schematic view of the cooler container of FIG. 1 on one embodiment of a charging base.

[0017] FIG. 3 is a partial view of the cooler container of FIG. 1, with a lid detached from the vessel of the cooler container, with three injector pens and/or cartridges coupled to the heat sink attached to the lid.

[0018] FIG. 4 is a schematic cross-sectional view of the cooler container of FIG. 1.

[0019] FIG. 5 is a schematic view of the cooler container of FIG. 1 in communication with a remote electronic device.

[0020] FIG. 6 is a schematic view of the cooler container of FIG. 1 and another embodiment of a charging base.

[0021] FIG. 7 is a schematic cross-sectional view of another embodiment of a cooler container.

[0022] FIG. 8 is a schematic cross-sectional view of a vessel of the cooler container of FIG. 7 without the lid.

[0023] FIG. 9 is a schematic block diagram showing communication between the cooler container and a remote electronic device.

[0024] FIG. 10A is a schematic partial perspective view of another cooler container.

[0025] FIG. 10B is a schematic cross-sectional view of the cooler container of FIG. 10A.

[0026] FIG. 11A is a schematic partial perspective view of another cooler container.

[0027] FIG. 11B is a schematic cross-sectional view of the cooler container of FIG. 11A.

[0028] FIG. 11C is a schematic cross-sectional view of the cooler container in FIG. 11A.

[0029] FIGS. 12A-12C is a schematic cross-sectional view of another cooler container.

[0030] FIG. 13 is a schematic partial cross-sectional view of a portion of another cooler container.

[0031] FIGS. 14A-14B are a schematic partial cross-sectional view of another cooler container.

[0032] FIG. 15 is a schematic partial cross-sectional view of another cooler container.

[0033] FIG. 16 shows a schematic perspective view of another cooler container and an exploded view of a capsule for use with the container.

[0034] FIG. 16A shows a schematic cross-sectional view of a capsule for use with the cooler container of FIG. 16.

[0035] FIG. 16B shows a schematic cross-sectional view of another capsule for use with cooler container of FIG. 16.

[0036] FIG. 16C shows an enlarged cross-sectional view of a portion of the capsule in FIG. 16B.

[0037] FIG. 17 shows a schematic perspective view of another cooler container.

[0038] FIG. 17A shows a schematic perspective view of a capsule for use with the cooler container of FIG. 17.

[0039] FIG. 17B shows a schematic cross-sectional view of the capsule in FIG. 17A for use with the cooler container of FIG. 17.

[0040] FIG. 18 shows a schematic perspective view of another cooler container.

[0041] FIG. 18A shows a schematic view of an injector pen for use with cartridges taken from the cooler container of FIG. 18.

[0042] FIG. 18B shows a schematic partial view of a cartridge from the cooler container of FIG. 18 loaded into an injector pen.

[0043] FIG. 19A shows a schematic perspective view of a cooler container.

[0044] FIG. 19B is a schematic block diagram showing electronics in the cooler container associated with operation of the display screen of the cooler container.

[0045] FIGS. 20A-20B show block diagrams of a method for operating the cooler container of FIG. 19A.

[0046] FIGS. 21A-21D are schematic user interfaces for an electronic device for use with a cooler container.

[0047] FIG. 22A is a schematic longitudinal cross-sectional view of a cooler container.

[0048] FIG. 22B is a schematic transverse cross-sectional view of the cooler container in FIG. 22A.

DETAILED DESCRIPTION

[0049] FIGS. 1-8 show a container system 100 (e.g., capsule container) that includes a cooling system 200. Optionally, the container system 100 has a container vessel 120 that is optionally cylindrical and symmetrical about a longitudinal axis Z, and one of ordinary skill in the art will recognize that the features shown in cross-section in FIGS. 4, 7 and 8 defined by rotating them about the axis Z to define the features of the container 100 and cooling system 200.

[0050] The container vessel 120 is optionally a cooler with active temperature control provided by the cooling system 200 to cool the contents of the container vessel 120 and/or maintain the contents of the vessel 120 in a cooled or chilled state. Optionally, the vessel 120 can hold therein one or more (e.g., a plurality of) separate containers 150 (e.g., medicine containers, such as injector pens, vials, cartridges (such as for injector pens), etc.). Optionally, the one or more (e.g., plurality of) separate containers 150 that can be inserted into the container vessel 120 can contain a medication or medicine (e.g., epinephrine, insulin, vaccines, etc.).

[0051] The container vessel 120 has an outer wall 121 that extends between a proximal end 122 that has an opening 123 and a distal end 124 having a base 125. The opening 123 is selectively closed by a lid L removably attached to the proximal end 122. As shown in FIG. 4, the vessel 120 has an inner wall 126A and a base wall 126B that together define an open chamber 126 that can receive and hold contents to be cooled therein (e.g., medicine containers, such as one or more vials, cartridges, injector pens, etc.). The vessel 120 can optionally have an intermediate wall 126C spaced about the inner wall 126A and base wall 126B, such that the intermediate wall 126C is at least partially disposed between the outer wall 121 and the inner wall 126A. The intermediate wall 126C is spaced apart from the inner wall 126A and base wall 126B so as to define a gap G between the intermediate wall 126C and the inner wall 126A and base wall 126B. The gap G can optionally be under vacuum so that the inner wall 126A and base wall 126B are vacuum insulated relative to the intermediate wall 126C and the outer wall 121 of the vessel 120.

[0052] Optionally, one or more of the inner wall 126A, intermediate wall 126B and outer wall 121 can be made of

metal (e.g., stainless steel). In one implementation, the inner wall 126A, base wall 126B and intermediate wall 126C are made of metal (e.g., stainless steel). In another implementation, one or more portions (e.g., outer wall 121, intermediate wall 126C and/or inner wall 126A) of the vessel 120 can be made of plastic.

[0053] The vessel 120 has a cavity 127 between the base wall 126B and a bottom 275 of the vessel 120. The cavity 127 can optionally house one or more batteries 277, and one or more printed circuit boards (PCBA) 278 with circuitry that controls the cooling system 200. In one implementation, the cavity 127 can optionally house a power button or switch actuable by a user through the bottom of the vessel 275, as further described below. Optionally, the bottom 275 defines at least a portion of an end cap 279 attached to the outer wall 121. Optionally, the end cap 279 is removable to access the electronics in the cavity 127 (e.g., to replace the one or more batteries 277, perform maintenance on the electronics, such as the PCBA 278, etc.). The power button or switch is accessible by a user (e.g., can be pressed to turn on the cooling system 200, pressed to turn off the cooling system 200, pressed to pair the cooling system 200 with a mobile electronic device, etc.). Optionally, the power switch can be located generally at the center of the end cap 279 (e.g., so that it aligns/extends along the longitudinal axis Z of the vessel 120).

[0054] With continued reference to FIGS. 1-8, the cooling system 200 is optionally at least partially housed in the lid L that releasably closes the opening 123 of the vessel 120. In one implementation, the lid L can releasably couple to the vessel 120 via one or more magnets in the lid L and/or in the vessel 120. In other implementations, the lid L can releasably couple to the vessel 120 via other suitable mechanisms (e.g., threaded connection, key-slot connection, press-fit connection, etc.).

[0055] In one implementation, the cooling system 200 can include a first heat sink (cold side heat sink) 210 in thermal communication with one or more thermoelectric elements (TECs) 220, such as Peltier element(s), and can be in thermal communication with the chamber 126 of the vessel 120 (e.g., via contact with the inner wall 126A, via conduction with air in the chamber 126, etc.). Optionally, cooling system 200 can include an insulator member (e.g., insulation material) disposed between the first heat sink 210 and a second heat sink 230.

[0056] With continued reference to FIGS. 1-8, the TEC 220 is selectively operated (e.g., by the circuitry 278) to draw heat from the first heat sink (e.g., cold-side heat sink) 210 and transfer it to the second heat sink (hot-side heat sink) 230. A fan 280 is selectively operable to draw air into the lid L to dissipate heat from the second heat sink 230, thereby allowing the TEC 220 to draw further heat from the first heat sink 210, and thereby draw heat from the chamber 126. During operation of the fan 280, intake air flow F_i is drawn through one or more intake vents 203 (having one or more openings 203A) in the lid L and over the second heat sink 230 (where the air flow removes heat from the second heat sink 230), after which the exhaust air flow F_o flows out of one or more exhaust vents 205 (having one or more openings 205A) in the lid L.

[0057] As shown in FIG. 4, the chamber 126 optionally receives and holds one or more (e.g., a plurality of) containers 150 (e.g., medicine containers, such as injector pens or cartridges for injector pens, vials, etc.). The first heat sink

210 can define one or more slots 211 that can receive and hold (e.g., resiliently receive and hold) one or more of the containers 150. Therefore, during operation of the cooling system 200, the first heat sink 210 is cooled, which thereby cools the one or more containers 150 coupled to the heat sink 210. In one implementation, the first heat sink 210 can be made of aluminum. However, the first heat sink 210 can be made of other suitable materials (e.g., metals with high thermal conductivity).

[0058] The electronics (e.g., PCBA 278, batteries 277) can electrically communicate with the fan 280 and TEC 220 in the lid L via one or more electrical contacts (e.g., electrical contact pads, Pogo pins) 281 in the lid L (e.g., downward facing electrical contacts, contact pads or Pogo pins) that contact one or more electrical contacts (e.g., Pogo pins, electrical contact pads) 282 in the portion of the vessel 120 (e.g., upward facing electrical contacts, contact pads or Pogo pins) that engages the lid L. Advantageously, the electrical contacts 281, 282 facilitate the coupling of the lid L to the vessel 120, 120' in the correct orientation (alignment) to allow the contact between the electrical contacts 282, 281 (e.g., provide a clocking feature). As shown in FIG. 3, the one or more electrical contacts 282 can be a set of eight contacts 282 that interface with an equal number of electrical contacts 281 in the lid L. However, different number of electrical contacts 282, 281 are possible. Electrical leads can extend from the PCBA 278 along the side of the vessel 120 (e.g., between the outer wall 121 and the intermediate wall 126C) to the electrical contacts 282. Accordingly, power can be provided from the batteries 277 to the TEC 220 and/or fan 280, and the circuitry (e.g., in or on the PCBA 278) can control the operation of the TEC 220 and/or fan 280, via one or more of the electrical contacts 281, 282 when the lid L is coupled to the vessel 120. As further discussed below, the lid L can have one or more sensors, and such sensors can communicate with the circuitry (e.g., in or on the PCBA 278) via one or more of the electrical contacts 281, 282.

[0059] FIGS. 7-8 schematically illustrate the container system 100 with the cooling system 200 and a vessel 120'. The cooling system 200 is similar to the cooling system 200 in the container 100 of FIGS. 1-7. Some of the features of the vessel 120' are similar to features in the vessel 120 in FIGS. 1-7. Thus, references numerals used to designate the various components of the vessel 120' are identical to those used for identifying the corresponding components of the vessel 120 in FIGS. 1-7, except that a "'" is added to the numerical identifier. Therefore, the structure and description for the various components of the cooling system 200 and vessel 120 in FIGS. 1-7 are understood to also apply to the corresponding components of the cooling system 200 and vessel 120' in FIGS. 7-8, except as described below.

[0060] As shown in FIGS. 7-8, the vessel 120' includes a cylindrical chamber wall 126D' that defines the chamber 126' and is spaced inward (e.g., toward the center of the chamber 126) of the inner wall 126A' and the base wall 126B' so as to define a gap G2' between the chamber wall 126D' and the inner wall 126A' and base wall 126B'. optionally, the gap G2' is filled with a phase change material (PCM) 130'. In one implementation, the phase change material 130' can be a solid-fluid PCM. In another implementation, the phase change material 130' can be a solid-solid PCM. The PCM 130' advantageously can passively absorb and release energy. Examples of possible PCM materials are water (which can transition to ice when cooled

below the freezing temperature), organic PCMs (e.g., bio based or Paraffin, or carbohydrate and lipid derived), inorganic PCMs (e.g., salt hydrates), and inorganic eutectics materials. However, the PCM 130' can be any thermal mass that can store and release energy.

[0061] In operation, the cooling system 200 can be operated to cool the heat sink 210 to cool the one or more containers 150 that are coupled to the heat sink 210, and to also cool the chamber 126'. The cooling system 200 can optionally also cool the PCM 130' (e.g., via the chamber wall 126D'). In one implementation, the cooling system 200 optionally cools the PCM 130' via conduction (e.g., contact) between at least a portion of the heat sink 210 and at least a portion of the chamber wall 126D' (e.g., near the opening 123' of the vessel 120'). In another implementation, the cooling system 200 optionally cools the PCM 130' via conduction through the air in the chamber 126' between the heat sink 210 and the chamber wall 126D'.

[0062] Advantageously, the PCM 130' operates as a secondary (e.g., backup) cooling source for the chamber 126' and/or the containers 150' (e.g., medicine containers, such as injector pens, cartridges for injector pens, vials, etc.) disposed in the chamber 126'. For example, if the one or more intake vents 203 are partially (or fully) blocked (e.g., because they are up against a surface of a handbag, backpack, suitcase, during travel; due to dust accumulation in the vent openings 203A) or if the cooling system 200 is not operating effectively due to low charge in the one or more batteries 277, the PCM 130' can maintain the one or more containers 150 (e.g., injector pens, cartridges for injector pens, vials, etc.) in a cooled state until the vents 203 are unblocked/unclogged, one or more batteries 277 are charged, etc. Though the phase change material 130' is described in connection with the chamber 126' and container system 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L one of skill in the art will recognize that it can also be applied to all the other implementations discussed herein for the chamber 126, 126', 126E, 126F1, 126F2, 126G1, 126H, 126I, 126J, 126K and container system 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L.

[0063] The container system 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L disclosed herein can optionally communicate (e.g., one-way communication, two-way communication) with one or more remote electronic devices (e.g., mobile phone, tablet computer, desktop computer, remote server) 600, via one or both of a wired or wireless connection (e.g., 802.11b, 802.11a, 802.11g, 802.11n standards, etc.). Optionally, the container system 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L can communicate with the remote electronic device 600 via an app (mobile application software) that is optionally downloaded (e.g., from the cloud) onto the remote electronic device 600. The app can provide one or more graphical user interface screens 610 via which the remote electronic device 600 can display one or more data received from the container system 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L and/or information transmitted from the remote electronic device 600 to the container system 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L. Optionally, a user can provide instructions to the container system 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L via the one or more of the graphical user interface screens 610 on the remote electronic device 600.

[0064] In one variation, the graphical user interface (GUI) screen 610 can provide one or more temperature presets corresponding to one or more particular medications (e.g., epinephrine/adrenaline for allergic reactions, insulin, vaccines, etc.). The GUI screen 610 can optionally allow the turning on and off of the cooling system 200, 200E, 200F, 200G, 200H, 200I, 200J, 200K, 200L. The GUI screen 610 can optionally allow the setting of the control temperature to which one or both of the first heat sink 210 and the chamber 126, 126', 126E, 126F1, 126F2, 126G1, 126H, 126I, 126J, 126K, 126L in the container 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L is cooled by the cooling system 200, 200E, 200F, 200G, 200H, 200I, 200J, 200K, 200L.

[0065] In another variation, the graphical user interface (GUI) screen 610 can provide a dashboard display of one or more parameters of the container 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L (e.g., ambient temperature, internal temperature in the chamber 126, 126', 126E, 126F1, 126F2, 126G1, 126H, 126I, 126J, 126K, 126L temperature of the first heat sink 210, temperature of the one or more batteries 277, etc.). The GUI screen 610 can optionally provide an indication (e.g., display) of power supply left in the one or more batteries 277 (e.g., % of life left, time remaining before battery power drains completely). Optionally, the GUI screen 610 can also include information (e.g., a display) of how many of the slots or receptacles 211 in the first heat sink 210 are occupied (e.g., by containers 150, 150J). Optionally, the GUI screen 610 can also include information on the contents of the container 100 (e.g., medication type, such as insulin, or disease medication is meant to treat, such as Hepatitis, etc.) and/or information (e.g., name, identification no., contact info) for the individual to whom the container 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L belongs.

[0066] In another variation, the GUI screen 610 can include one or more notifications provided to the user of the container system 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L disclosed herein, including alerts on battery power available, alerts on ambient temperature effect on operation of container system 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L alert on temperature of the first heat sink 210, alert on temperature of the chamber 126, 126', 126E, 126F, 126G, 126H, 126I, 126J, 126K, 126L alert on low air flow through the intake vent 203 and/or exhaust vent 205 indicating they may be blocked/clogged, etc. One of skill in the art will recognize that the app can provide the plurality of GUI screens 610 to the user, allowing the user to swipe between the different screens. Optionally, as discussed further below, the container system 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K can communicate information, such as temperature history of the chamber 126, 126', 126E, 126F, 126G, 126H, 126I, 126J, 126K, 126L temperature history of the first heat sink 210 and/or chamber 126, 126', 126E, 126F, 126G, 126H, 126I, 126J, 126K, 126L that generally corresponds to the temperature of the containers 150, 150J, temperature of the container 150, 150J from a temperature sensor on the container 150, 150J, power level history of the batteries 277, ambient temperature history, etc. to one or more of a) an RFID tag on the container system 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L that can later be read (e.g., at the delivery location), b) to a remote electronic device (e.g., a mobile electronic device such as a smartphone or tablet computer or laptop computer or desktop computer), including wirelessly (e.g., via WiFi

802.11, BLUETOOTH®, or other RF communication), and c) to the cloud (e.g., to a cloud-based data storage system or server) including wirelessly (e.g., via WiFi 802.11, BLUETOOTH®, or other RF communication). Such communication can occur on a periodic basis (e.g., every hour; on a continuous basis in real time, etc.). Once stored on the RFID tag or remote electronic device or cloud, such information can be accessed via one or more remote electronic devices (e.g., via a dashboard on a smart phone, tablet computer, laptop computer, desktop computer, etc.). Additionally, or alternatively, the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** can store in a memory (e.g., part of the electronics in the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L**) information, such as temperature history of the chamber **126**, **126'**, **126E**, **126F**, **126G**, **126H**, **126I**, **126J**, **126K**, **126L** temperature history of the first heat sink **210**, power level history of the batteries **277**, ambient temperature history, etc., which can be accessed from the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** by the user via a wired or wireless connection (e.g., via the remote electronic device **600**).

[0067] With reference to FIGS. 1-9, the body **120** of the container **100** can optionally have a visual display on the outer wall **121** of the body **120**. The visual display can optionally display one or more of the temperature in the chamber **126**, **126'**, the temperature of the first heat sink **210**, the ambient temperature, a charge level or percentage for the one or more batteries **277**, and amount of time left before recharging of the batteries **277** is needed, etc. The visual display can optionally include a user interface (e.g., pressure sensitive buttons, capacitance touch buttons, etc.) to adjust (up or down) the temperature preset at which the cooling system **200** is to cool the chamber **126**, **126'**. Accordingly, the operation of the container **100** (e.g., of the cooling system **200**) can be selected via the visual display and user interface on a surface of the container **100**. Optionally, the visual display can include one or more hidden-til-lit LEDs. Optionally, the visual display can include an electronic ink (e-ink) display. In one variation, the container **100** can optionally include a hidden-til-lit LED **140** that can selectively illuminate (e.g., to indicate one or more operating functions of the container **100**, such as to indicate that the cooling system **200** is in operation). The LED **140** can optionally be a multi-color LED selectively operable to indicate one or more operating conditions of the container **100** (e.g., green if normal operation, red if abnormal operation, such as low battery charge or inadequate cooling for sensed ambient temperature, etc.). Though the visual display is described in connection with the container system **100**, one of skill in the art will recognize that it can also be applied to all the other implementations discussed herein for the container system **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L**.

[0068] In operation, the cooling system **200** can optionally be actuated by pressing a power button. Optionally, the cooling system **200** can additionally (or alternatively) be actuated remotely (e.g., wirelessly) via a remote electronic device **600**, such as a mobile phone, tablet computer, laptop computer, etc. that wirelessly communicates with the cooling system **200** (e.g., with a receiver or transceiver of the circuitry **278**). In still another implementation, the cooling system **200** can automatically cool the chamber **126**, **126'** when the lid **L** is coupled to the vessel **120**, **120'** (e.g., upon

receipt by the circuitry, for example in or on the PCBA **278**, of a signal, such as from a pressure sensor, proximity sensor, load sensor, light sensor) that the lid **L** has been coupled with the vessel **120**, **120'**. The chamber **126**, **126'** can be cooled to a predetermined and/or a user selected temperature or temperature range, or automatically cooled to a temperature preset corresponding to the contents in the containers **150** (e.g., insulin, epinephrine, vaccines, etc.). The user selected temperature or temperature range can be selected via a user interface on the container **100** and/or via the remote electronic device **600**.

[0069] The circuitry **278** optionally operates the one or more TECs **220** so that the side of the one or more TECs **220** adjacent the first heat sink **210** is cooled to thereby cool the one or more containers **150** in thermal communication with (e.g., coupled to) the first heat sink **210** and so that the side of the one or more TECs **220** adjacent the one or more second heat sinks **230** is heated. The TECs **220** thereby cool the first heat sink **210** and thereby cools the containers **150** and/or the chamber **126**, **126'**. The container **100** can include one or more sensors (e.g., temperature sensors) **155** operable to sense a temperature of the chamber **126**, **126'**. As best shown in FIG. 7, the one or more sensors **155** can include a temperature sensor that extends through one or more of the prongs of the first heat sink **210** and protrudes from the first heat sink **210** into the chamber **126**, **126'** when the lid **L** is coupled to the vessel **120**, **120'**. The one or more sensors **155** can communicate information to the circuitry **278** indicative of the sensed temperature(s) via the one or more electrical contacts **281**, **282** when the lid **L** is coupled to the vessel **120**, **120'**. The circuitry (e.g., in or on the PCBA **278**) operates one or more of the TECs **220** and one or more fans **280** based at least in part on the sensed temperature information (from the one or more sensors **155**) to cool the first heat sink **210** and/or the chamber **126**, **126'** to the predetermined temperature (e.g., temperature preset) and/or user selected temperature. The circuitry operates the one or more fans **280** to flow air (e.g., received via the intake vents **203**) over the one or more second heat sinks **230** to dissipate heat therefrom, thereby allowing the one or more second heat sinks **230** to draw more heat from the one or more TECs **220**, which in turn allows the one or more TEC's **220** to draw more heat from (i.e., cool) the first heat sink **210** and optionally the chamber **126**, **126'**. Said air flow, once it passes over the one or more second heat sinks **230**, is exhausted via the exhaust vents **205**.

[0070] With reference to FIG. 2, a power base **300** can receive the container **100** thereon and can provide power to the electronics in the container **100** to, for example, charge the one or more batteries **277** or provide power directly to the TECs **220** and/or fan **280**. In one implementation, the power base **300** has an electrical cord that ends in an electrical connector (wall plug, USB connector), which allows the power base **300** to connect to a power source (e.g., wall outlet, USB connector of power source, such as a laptop or desktop computer). In one implementation, the power base **300** transmits power to the container **100** via inductive coupling. In another implementation, the power base **300** transmits power to the container **100** via one or more electrical contacts (e.g., electrical contact pads, Pogo pins) that contact one or more electrical contacts (e.g., electrical contact pads, contact rings) on the container **100** (e.g., on the bottom **275** of the container **100**).

[0071] FIG. 6 shows a power base 300' that can receive the container 100 thereon and can provide power to the electronics in the container 100 to, for example, charge the one or more batteries 277 or provide power directly to the TEC 220 and/or fan 280. The power base 300' is similar to the power base 300 except as described below. In one implementation, the power base 300' has an electrical cord that ends in an electrical connector (for a car charger), which allows the power base 300' to connect to a car charger. Advantageously, the power base 300' is sized to fit in a cup holder of an automobile, allowing the container 100 to be placed in the cupholder while on the power base 300', keeping the container 100 in a substantially stable upright orientation.

[0072] In one variation, the container system 100 is powered using 12 VDC power (e.g., from one or more batteries 277 or power base 300'). In another variation, the container system 100 is powered using 120 VAC or 240 VAC power, for example using the power base 300. The circuitry 278 in the container 100 can include a surge protector to inhibit damage to the electronics in the container 100 from a power surge.

[0073] FIG. 9 shows a block diagram of a communication system for (e.g., incorporated into) the devices described herein (e.g., the one or more container systems 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L). In the illustrated embodiment, circuitry EM (e.g., on the PCBA 278) can receive sensed information from one or more sensors S1-Sn (e.g., level sensors, volume sensors, temperature sensors, such as sensors 155, battery charge sensors, biometric sensors, load sensors, Global Positioning System or GPS sensors, radiofrequency identification or RFID reader, etc.). The circuitry EM can be housed in the container, such as in the vessel 120, 120', 120E, 120F, 120G, 120H, 120I, 120J, 120K (e.g., bottom of vessel 120, 120', 120E, 120F, 120G, 120H, 120I, 120J, 120K, 120L side of vessel 120, 120', 120E, 120F, 120G, 120H, 120I, 120J, 120K, 120L as discussed above) or in a lid L of the container. The circuitry EM can receive information from and/or transmit information (e.g., instructions) to one or more heating or cooling elements HC, such as the TEC 220, 220E, 220F1, 220F2, 220G, 220L (e.g., to operate each of the heating or cooling elements in a heating mode and/or in a cooling mode, turn off, turn on, vary power output of, etc.) and optionally to one or more power storage devices PS (e.g., batteries 277, 277E, 277F, 277L such as to charge the batteries or manage the power provided by the batteries to the one or more heating or cooling elements 220, 220E, 220F1, 220F2, 220G, 220L).

[0074] Optionally, the circuitry EM can include a wireless transmitter, receiver and/or transceiver to communicate with, e.g., transmit information, such as sensed temperature, position data, to and receive information, such as user instructions, from one or more of: a) a user interface UI1 on the unit (e.g., on the body of the vessel 120, 120E, 120F, 120G, 120H, 120I, 120J, 120K, 120L), b) an electronic device ED (e.g., a mobile electronic device such as a mobile phone, PDA, tablet computer, laptop computer, electronic watch, a desktop computer, remote server), c) the cloud CL (e.g., a cloud-based data storage system), or d) communicate via a wireless communication system such as WiFi and Bluetooth BT. The electronic device ED (such as electronic device 600) can have a user interface UI2 (such as GUI 610), that can display information associated with the operation of the container system, and that can receive information (e.g.,

instructions) from a user and communicate said information to the container system 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L (e.g., to adjust an operation of the cooling system 200, 200E, 200F, 200G, 200H, 200I, 200J, 200K, 200L).

[0075] In operation, the container system 100 can operate to maintain one or both of the first heat sink 210 and the chamber 126, 126' of the vessel 120, 120' at a preselected temperature or a user selected temperature. The cooling system 200 can operate the one or more TECs 220 to cool the first heat sink 210 and, optionally the chamber 126, 126', 126E, 126F1, 126F2, 126G1, 126L (e.g., if the temperature of the first heat sink 210 or chamber 126, 126', 126E, 126F1, 126F2, 126G1, 126L is above the preselected temperature, such as when the ambient temperature is above the preselected temperature) or to heat the first heat sink 210 and, optionally chamber 126, 126', 126E, 126F1, 126F2, 126G1, 126L (e.g., if the temperature of the first heat sink 210 or chamber 126, 126', 126E, 126F1, 126F2, 126G1, 126L is below the preselected temperature, such as when the ambient temperature is below the preselected temperature). The preselected temperature may be tailored to the contents of the container (e.g., a specific medication, a specific vaccine, insulin pens, epinephrine pens or cartridges, etc.), and can be stored in a memory of the container 100, and the cooling system 200 or heating system, depending on how the temperature control system is operated, can operate the TEC 220 to approach the preselected or set point temperature.

[0076] Optionally, the circuitry EM can communicate (e.g., wirelessly) information to a remote location (e.g., cloud based data storage system, remote computer, remote server, mobile electronic device such as a smartphone or tablet computer or laptop or desktop computer) and/or to the individual carrying the container (e.g., via their mobile phone, via a visual interface on the container, etc.), such as a temperature history of the first heat sink 210, 210E1, 210E2, 210F1, 210F2, 210L and/or chamber 126, 126' 126E, 126F1, 126F2, 126G1, 126L to provide a record that can be used to evaluate the efficacy of the medication in the container and/or alerts on the status of the medication in the container 100, 100E, 100F, 100G, 100H, 100I, 100J, 100K, 100L. Optionally, the temperature control system (e.g., cooling system, heating system) 200, 200E, 200F, 200G, 200H, 200I, 200J, 200K, 200L automatically operates the TEC 220, 220E, 220F1, 220F2, 220L to heat or cool the first heat sink 210, 210E1, 210E2, 210F1, 210F2, 210L and, optionally, the chamber 126, 126', 120E, 120F1, 210F2 of the vessel 120, 120', 120E, 120F to approach the preselected temperature. In one implementation, the cooling system 200, 200E, 200F, 200G, 200H, 200I, 200J, 200K, 200L can cool and maintain one or both of the chamber 126, 126', 126E, 126F1, 126F1, 126G1, 126L and the containers 150 at or below 15 degrees Celsius, such as at or below 10 degrees Celsius, in some examples at approximately 5 degrees Celsius.

[0077] In one implementation, the one or more sensors S1-Sn can include one more air flow sensors in the lid L that can monitor airflow through one or both of the intake vent 203 and exhaust vent 205. If said one or more flow sensors senses that the intake vent 203 is becoming clogged (e.g., with dust) due to a decrease in air flow, the circuitry EM (e.g., on the PCBA 278) can optionally reverse the operation of the fan 280, 280E, 280F for one or more predetermined periods of time to draw air through the exhaust vent 205 and

exhaust air through the intake vent **203** to clear (e.g., unclog, remove the dust from) the intake vent **203**. In another implementation, the circuitry **EM** can additionally or alternatively send an alert to the user (e.g., via a user interface on the container **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L**, wirelessly to a remote electronic device such as the user's mobile phone via GUI **610**) to inform the user of the potential clogging of the intake vent **203**, so that the user can inspect the container **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L** and can instruct the circuitry **EM** (e.g., via an app on the user's mobile phone) to run an "cleaning" operation, for example, by running the fan **280**, **280E**, **280F** in reverse to exhaust air through the intake vent **203**.

[0078] In one implementation, the one or more sensors **S1-Sn** can include one more Global Positioning System (GPS) sensors for tracking the location of the container system **100**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100L**. The location information can be communicated, as discussed above, by a transmitter and/or transceiver associated with the circuitry **EM** to a remote location (e.g., a mobile electronic device, a cloud-based data storage system, etc.).

[0079] In another variation, the circuitry **278** and one or more batteries **277** can be in a removable pack (e.g., DeWalt battery pack) that attaches to the distal end **124** of the vessel **120**, **120'**, **120E**, **120F**, where one or more contacts in the removable pack contact one or more contacts on the distal end **124** of the vessel **120**, **120'**, **120E**, **120F**, **120G**. The one or more contacts on the distal end **124** of the vessel **120**, **120'**, **120E**, **120F**, **120G** are electrically connected (via one or more wires or one or more intermediate components) with the electrical connections on the proximal end **122** of the vessel **120**, **120E**, **120F**, **120G**, **120H**, **120I**, **120J**, **120K**, or via as discussed above, to provide power to the components of the cooling system **200**, **200E**, **200F**, **200G**, **200H**, **200I**, **200J**, **200K**, **200L**.

[0080] FIGS. 10A-10B show a container system **100E** (e.g., capsule container) that includes a cooling system **200E**. The container system **100E** and cooling system **200E** are similar to the container system **100** and cooling system **200** described above in connection with FIGS. 1-8. Thus, references numerals used to designate the various components of the container vessel **100E** and cooling system **200E** are identical to those used for identifying the corresponding components of the container system **100** and cooling system **200** in FIGS. 1-8, except that an "E" is added to the numerical identifier. Therefore, the structure and description for the various components of the container system **100** and cooling system **200** in FIGS. 1-8 is understood to also apply to the corresponding components of the container system **100E** and cooling system **200E** in FIGS. 10A-10B, except as described below.

[0081] The container system **100E** differs from the container system **100** in that the opening **123E** in the vessel **120E** has an oval shape and the open chamber **126E** has an oval cross-section. The chamber **126E** is sized to receive a pair of containers **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) side-by-side therein. The container **100E** has electrical contacts **282E** that can interface with electrical contacts **281E** in the lid **L**.

[0082] The lid **L** can have a pair of spaced apart plates **211E1**, **211E2** that can hold the pair of containers (e.g.,

medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) therebetween, such as in slots between the plates **211E1**, **211E2**. The plates **211E1**, **211E2** can be part of the first heat sink **210E** in thermal communication with one or more TECs **220E**, such as Peltier element(s), and be in thermal communication with the chamber **126E** of the vessel **120E** (when the lid **L** is attached to the vessel **120E**). As shown in FIG. 10B, the plates **211E1**, **211E2** can be interposed between the containers **150** (medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) and the inner wall **126AE** of the chamber **126E**.

[0083] The chamber **126E** can be approximately $\frac{1}{2}$ as large as the chamber **126** of vessel **120** (which is sized to hold up to four containers **150**). The other half of the vessel **100E** can house one or more batteries **277E** therein. The chamber **126E** can be insulated (e.g., vacuum insulated) relative to the outer wall **121E** of the vessel **120E**.

[0084] FIGS. 11A-11C show a container system **100F** (e.g., capsule container) that includes a cooling system **200F**. The container system **100F** and cooling system **200F** are similar to the container system **100** and cooling system **200** described above in connection with FIGS. 1-8. Thus, references numerals used to designate the various components of the container system **100F** and cooling system **200F** are identical to those used for identifying the corresponding components of the container system **100** and cooling system **200** in FIGS. 1-8, except that an "F" is added to the numerical identifier. Therefore, the structure and description for the various components of the container system **100** and cooling system **200** in FIGS. 1-8 is understood to also apply to the corresponding components of the container vessel **100F** and cooling system **200F** in FIGS. 11A-11C, except as described below.

[0085] The container system **100F** differs from the container system **100** in that the vessel **120F** has two openings **123F1**, **123F2** at the top of two separate and spaced apart chambers **126F1**, **126F2**. Optionally, the openings **123F1**, **123F2** has a circular shape and each of the chambers **126F1**, **126F2** has a circular cross-section. Each of the chambers **126F1**, **126F2** is sized to receive a container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) side-by-side therein. The container vessel **100F** has two separate groups of electrical contacts **282F1**, **282F2** that can interface with electrical contacts **281F1**, **281F2** in the lid **L**.

[0086] The lid **L** can have a pair of spaced apart heat sinks **210F1**, **210F2**, each sized to resiliently hold one container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.), for example in a slot defined by the heat sinks **210F1**, **210F2**. Each of the heat sinks **210F1**, **210F2** can be in thermal communication with a separate TEC **220F1**, **220F2**, which in turn can optionally be in thermal communication with separate second heat sinks (not shown) in the lid **L**. As discussed in FIGS. 1-8, the cooling system **200F** can have one or more fans **280F** operable to draw air over the second heat sinks (not shown) in the lid **L**. The chambers **126F1**, **126F2** can be insulated (e.g., vacuum insulated) relative to each other and relative to the outer wall **121F** of the vessel **100F**.

[0087] Advantageously, the heat sinks **210F1**, **210F2** can be operated independently of each other. Accordingly, in one implementation both heat sinks **210F1**, **210F2** are operable to cool the containers **150** to the approximately the same

temperature (e.g., down to approximately 5 degrees Celsius) when the containers **150** are in the chambers **126F1**, **126F2** and the lid **L** is disposed on top of the vessel **120F** to seal the vessel **120F**. In another implementation both heat sinks **210F1**, **210F2** are operable to cool the containers **150** to different temperatures when the containers **150** are in the chambers **126F1**, **126F2** and the lid **L** is disposed on top of the vessel **120F** to seal the vessel **120F**. In another implementation, for example when a user is ready or almost ready to consume the medicine in the container **100F**, one of the heat sinks **210F1** can be heated to heat its associated container **150** (e.g., to a predetermined consumption or administration temperature, for example to body temperature, to room temperature), while the other heat sink **210F2** cools its associated container **150** in the associated chamber **126F2**. In still another implementation, both heat sinks **210F1**, **210F2** are operated to heat their associated containers **150** (e.g., to the same temperature, to different temperatures).

[0088] FIGS. 12A-12C show a container system **100G** (e.g., a capsule container) that includes a cooling system **200G**. The container system **100G** and cooling system **200G** are similar to the container system **100F** and cooling system **200F** described above in connection with FIGS. 11A-11C. Thus, references numerals used to designate the various components of the container system **100G** and cooling system **200G** are identical to those used for identifying the corresponding components of the container system **100F** and cooling system **200F** in FIGS. 11A-11C, except that a “G” instead of an “F” is added to the numerical identifier. Therefore, the structure and description for the various components of the container system **100F** and cooling system **200F** in FIGS. 11A-11C is understood to also apply to the corresponding components of the container vessel **100G** and cooling system **200G** in FIGS. 12A-12C, except as described below. For clarity, FIG. 12A only shows one chamber **126G1**, but can have two chambers **126G1**, **126G2** similar to chambers **126F1**, **126F2** described above. Optionally, the chamber(s) **126G1**, **126G2** are removable from the container system **100G**, as further described below.

[0089] The container system **100G** differs from the container system **100F** in that the heat sink **210G1** is a removable sleeve **210G1** that removably couples to the container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.). The sleeve **210G1** can be made of a thermally conductive material (e.g., a metal, such as aluminum). The sleeve **210G1** can be removed along with the container **150** from the container vessel **120G** (e.g., for placement in a user’s purse, backpack, work bag during a commute or travel, etc.). Optionally, the sleeve **210G1** can maintain the container **150** in a cooled state for an extended period of time (e.g., between about 1 hour and about 10 hours, between about 1 hour and about 5 hours, between about 1 hour and about 3 hours, about 2 hours, etc.). When the sleeve **210G1** is coupled with the container **150** and inserted into the chamber **126G1**, the sleeve **210G1** can interface with the cooling system **200G** and operate as a heat transfer interface between the cooling system **200G** (e.g., between one or more TECs **220G** of the cooling system **200G** and the container **150**) to help cool and/or heat the container **150**. For example, when the cooling system **200G** is used to cool the container **150**, the

sleeve **210G1** can function as a heat sink to remove heat (e.g., cool) the container **150** that is attached to the sleeve **210G1**.

[0090] With reference to FIG. 12C, the sleeve **210G1** can have a top surface **210G2**, an outer wall **210G3** and an inner wall **210G4**, where at least a portion of the inner wall **210G4** can be in contact with the container **150** when the sleeve **210G1** is coupled to the container **150**. Optionally, the sleeve **210G1** can define a cavity (e.g., an annular cavity) **210G5** between the outer wall **210G3** and the inner wall **210G4**. In one implementation, the cavity **210G5** can house a thermal mass material **130G**. In one implementation, the thermal mass material **130G** is a phase change material PCM (e.g., a solid-solid PCM, a solid-fluid PCM) that can transition from a heat absorbing state to a heat releasing state at a transition temperature. In another implementation, the cavity **210G5** is excluded and the sleeve **210G1** instead has a wall that extends between the inner surface **210G4** and the outer wall **210G3** with a thermal surface that can absorb and release heat.

[0091] The sleeve **210G1** can optionally include a heater **210G6** (e.g., a flex heater) in thermal communication with the inner wall **210G4** (e.g., the heater **210G6** can be disposed on the inner wall **210G4**, embedded in the inner wall **210G4**, disposed behind the inner wall **210G4** (e.g., disposed in the cavity **210G5**)). The sleeve **210G1** can have one or more electrical contacts **210G7** on a surface thereof (e.g., on the top surface **210G2**). The one or more electrical contacts **210G7** can be in electrical communication with the heater **210G6**. In another implementation, the sleeve **210G1** can exclude the heater **210G6** and one or more electrical contacts **210G7**.

[0092] In operation, while the sleeve **210G1** is coupled to the container **150** and inserted into the container vessel **120G** with the lid **L** in the closed position relative to the container vessel **120G**, the cooling system **200G** can operate to cool one or both of the chamber **126G1** and the sleeve **210G1**. For example, one or more TECs **220G** of the cooling system **200G** can cool a heat sink surface that contacts the top surface **210G2** of the sleeve **210G1**, thereby also being placed in thermal communication with the inner wall **210G4**, outer wall **210G3** and optional thermal mass **130G** (e.g., PCM) in the cavity **210G5**. The TECs **220G** can thereby cool the sleeve **210G1** and thereby cool the container **150** attached to it, as well as charge the optional thermal mass **130G** (e.g., PCM). Optionally, where the sleeve **210G1** includes the heater **210G6**, a controller of the system **200G** can operate the heater **210G6** to heat the contents of the container **150** (e.g., to room temperature, body temperature) prior to the container **150** being removed from the container vessel **120G** for use (e.g. for application of the contents of the container to the user, such as via an injector pen). For example, the controller can provide power to the heater **210G6** via the electrical contacts **210G7** that contact electrical contacts in the lid **L** when the lid **L** is in a closed position relative to the container vessel **100**.

[0093] In one implementation, once the cooling system **200G** has cooled the sleeve **210G1** and its attached container **150**, the user can optionally remove the sleeve **210G1** with its attached container **150** from the container vessel **120G**, as described above (e.g., for travel, commute, etc.) and the charged thermal mass **130G** can maintain the container **150** attached to the sleeve **210G1** in a cooled state for an extended period of time, as discussed above.

[0094] FIG. 13 shows another implementation of a chamber 126G1 in the container system 100G (e.g., a capsule container) that includes a cooling system 200G. As discussed above, the chamber 126G1 can receive a container 150 (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) attached to the sleeve 210G1. The chamber 126G1 can be actuated between a retracted position and an extended position in the container vessel 100G. As shown in FIG. 13, the chamber 126G1 can be spring loaded within the container vessel 100G. A guide 430 can guide the movement of the chamber 126G1 between the retracted and extended position.

[0095] In one implementation, the chamber 126G1 can have an actuation mechanism 400 that can optionally include a spring 410 that extends between a bottom of the chamber 126G1 and a cam 420. The spring 410 can be a compression spring. In one implementation, the cam 240 can move between a first orientation to position the chamber 126G1 in the retracted position and a second orientation to position the chamber 126G2 in the extended position. The movement of the cam 240 to change its orientation can be actuated by pushing down on the sleeve 210G1 (e.g., on the top surface 210G2 of the sleeve 210G1). Movement of the chamber 126G1 to the extended position can facilitate removal of the container 150 (e.g., with the attached sleeve 210G1) from the chamber 126G1 (e.g., when ready for use by the user, as discussed above).

[0096] Optionally, with the chamber 126G1 in the extended position, and with the container 150 in the chamber 126G1 and attached to the sleeve 210G1, movement of the lid L to the closed position relative to the container vessel 120G can urge the chamber 126G1 into the container vessel 120G and actuate the movement of the cam 420 to allow the chamber 126G1 to move to the retracted position. Though the actuation mechanism 400 is described in connection with the chamber 126G1 and container system 100G, one of skill in the art will recognize that the features of the actuation mechanism 400 described herein can also be applied to all the other implementations discussed herein for the container system 100, 100E, 100F, 100G.

[0097] FIGS. 14A-14B shows another implementation of a chamber 126G1 in the container system 100G (e.g., a capsule container) that includes a cooling system 200G. As discussed above, the chamber 126G1 can receive a container 150 (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) attached to the sleeve 210G1. The chamber 126G1 can be actuated between a retracted position and an extended position in the container vessel 120G. As shown in FIGS. 14A-14B, the chamber 126G1 can be actuated between the retracted position and the extended position by an actuation mechanism 400'. The actuation mechanism 400' can optionally be housed in the container vessel 120G below the chamber 126G1 (e.g., between a bottom of the chamber 126G1 and a bottom of the container vessel 120G). A guide 430 can guide the movement of the chamber 126G1 between the retracted and extended position.

[0098] With reference to FIG. 14B, the actuation mechanism 400' can include a linear actuator 410' and a motor 420' operable to drive the linear actuator 410'. The linear actuator 410' can optionally include a coupling that couples to an output shaft of the motor 420'. The coupling 412' is coupled to a ball screw 414' that rotates when the motor 420' rotates the coupling 412'. The ball screw 414' rotates relative to a

ball screw nut 416', where the ball screw nut 416' travels along the ball screw 414' as the motor 420' rotates the coupling 412' (e.g., travels rightward in the drawing when coupling 412' rotates in one direction and travels leftward in the drawing when the coupling 412' rotates in the opposite direction). The ball screw nut 416' can be attached to a rod such that the rod translates (at least partially within a bushing 419') along the axis of the ball screw 414' as the screw 414' rotates. An end of the rod 418' can engage a bottom of the chamber 126G1 to move the chamber 126G1 between the retracted and extended position relative to the container vessel 120G, however, in other implementations, the actuation mechanism 400' can be other suitable linear motion mechanisms (e.g., instead of an electric motor 420' can include a pneumatic or hydraulic system to translate the rod 418'). Though the actuation mechanism 400' is described in connection with the chamber 126G1 and container vessel 120G, one of skill in the art will recognize that the features of the actuation mechanism 400' described herein can also be applied to all the other implementations discussed herein for the container vessel 100, 100E, 100F, 100G.

[0099] FIG. 15 shows another implementation of a chamber 126G1 in the container system 100G (e.g., a capsule container) that includes a cooling system 200G. As discussed above, the chamber 126G1 can receive a container 150 (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) attached to the sleeve 210G1. The chamber 126G1 can be actuated between a retracted position and an extended position in the container vessel 120G. As shown in FIG. 15, the chamber 126G1 can be actuated between the retracted position and the extended position by an actuation mechanism 400". The actuation mechanism 400" can optionally be housed in the lid L. Though not shown, a guide (similar to guide 430) can guide the movement of the chamber 126G1 between the retracted and extended position.

[0100] With reference to FIG. 15, the actuation mechanism 400" can include a magnet 420". In one implementation, the magnet 420" can be an electromagnet. In operation, the electromagnet 420" can be operated to draw the sleeve 210G1 (e.g., the top surface 210G2 of the sleeve 210G1) into contact with a heat sink surface and/or one or more TECs 220G to place the sleeve 210G1 (and therefore the container 150 coupled to the sleeve 210G1) in thermal communication with the one or more TECs 220G, which can be operated to cool the sleeve 210G1 and/or container 150 and/or the chamber 126G1. The electromagnet 420" can be turned off or not operated to allow the sleeve 210G1 (and container 150 attached to it) to be displaced from the heat sink and/or one or more TECs 220G to thereby thermally disconnect the container 150 and sleeve 210G1 from the TECs 220G. The electromagnet 420" can be turned off or disengaged when, for example, the user wishes to remove the container 150 and sleeve 210G1 from the container vessel 120G (e.g., for storing in another compartment, such as a purse, backpack, travel bag, etc. during a commute or trip). Though the actuation mechanism 400" is described in connection with the chamber 126G1 and container vessel 120G, one of skill in the art will recognize that the features of the actuation mechanism 400" described herein can also be applied to all the other implementations discussed herein for the container vessel 100, 100E, 100F, 100G.

[0101] In another implementation, the container system 100, 100E, 100F, 100G can have chambers 126, 126E,

126F1, 126F2, 126G1 that can be completely removed from the container vessel **120**, **120E**, **120F**, **120F**, such as for travel or commute, where the chamber can hold the container **150** (e.g., vial, cartridge (such as for use in injector pen), injector pen, etc.) therein (e.g., provide a travel pack) until the container **150** is ready for use.

[0102] FIGS. **16A-16C** show a container system **100H** (e.g., a capsule container) that includes a cooling system **200H**. The container system **100H** and cooling system **200H** are similar to the container system **100G** and cooling system **200G** described above in connection with FIGS. **12A-12C**. Thus, references numerals used to designate the various components of the container system **100H** and cooling system **200H** are identical to those used for identifying the corresponding components of the container system **100G** and cooling system **200G** in FIGS. **12A-12C**, except that an “H” instead of a “G” is added to the numerical identifier. Therefore, the structure and description for the various components of the container system **100G** and cooling system **200G** in FIGS. **12A-12C** is understood to also apply to the corresponding components of the container system **100H** and cooling system **200H** in FIGS. **16A-16C**, except as described below.

[0103] As shown in FIG. **16**, the container system **100H** has a container vessel **120H** and a lid **L**. The lid **L** can include a cooling system **200G**. The container vessel **120H** can optionally have one or more chambers **126H** that extend to corresponding one or more openings **123H**. Though FIG. **16** shows the container vessel **120H** having six chambers **126H**, one of skill in the art will recognize that the container vessel **120H** can have more or fewer chambers **126H** than shown in FIG. **16**. The chamber(s) **126H** of the container vessel **120H** can removably hold a corresponding capsule **210H** therein. In one implementation, the container vessel **120H** can have the same or similar structure as shown and described above for the container vessel **120**, **120E**, **120F**, **120G**. Optionally, the container vessel **120H** can have a cavity between the chamber(s) **126H** and the outer surface of the container vessel **120H** that is vacuum insulated. In another implementation, the container vessel **120H** excludes vacuum insulation and can instead have a gap or cavity between the chamber(s) **126H** and an outer surface of the container vessel **120H** that is filled with air. In still another implementation, the container vessel **120H** can have a gap or cavity between the chamber(s) **126H** and an outer surface of the container vessel **120H** that includes an insulating material.

[0104] With continued reference to FIG. **16**, the capsule(s) **210H** have a vessel portion **210H1** and a lid portion **210H2** that together can enclose a container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.). The lid portion **210H2** can be moved between a closed position relative to (e.g., adjacent) the vessel portion **210H1** and an open position relative to (e.g., spaced apart from) the vessel portion **210H1**. In the closed position, the lid portion **210H2** can optionally be held against the vessel portion **210H1** (e.g., by one or more magnetic surfaces of the lid portion **210H2** and/or vessel portion **210H1**) to inhibit (e.g., prevent) the container **150** from inadvertently falling out of the capsule **210H**.

[0105] FIG. **16A** shows one implementation of a capsule **210H**, where the vessel portion **210H1** and lid portion **210H2** have an outer surface **210H3** and an inner surface **210H4** that defines a cavity **210H8** that receives the con-

tainer **150**. The vessel portion **210H1** and lid portion **210H2** can also have one or more intermediate walls **210H6** radially between the inner surface **210H4** and the outer surface **210H3** that define a first cavity **210H5** between the inner wall **210H4** and the intermediate wall(s) **210H6** and a second cavity **210H9** between the intermediate wall(s) **210H6** and the outer surface **210H3**. Optionally, the second cavity **210H5** can be vacuum insulated (i.e., the second cavity **210H5** can be under vacuum or negative pressure force). Optionally, the first cavity **210H5** can house a thermal mass material **130H**. In one implementation, the thermal mass material **130H** is a phase change material PCM (e.g., a solid-solid PCM, a solid-fluid PCM) that can transition from a heat absorbing state to a heat releasing state at a transition temperature. In another implementation, the cavity **210H5** is excluded and the capsule **210H** instead has a wall that extends between the inner surface **210H4** and the intermediate wall(s) **210H6** that can absorb and release heat.

[0106] With continued reference to FIG. **16A**, the capsule **210H** has a thermally conductive contact **210H7** at one or both ends of the capsule **210H**. The thermally conductive contact **210H7** can be made of metal, though it can be made of other thermally conductive material. In one implementation, the thermally conductive contact **210H7** is made of copper. The thermally conductive contact **210H** can extend from the outer surface **210H3** to the inner surface **210H4** and through the first and second cavities **210H5**, **210H9**, so as to be in thermal contact with the thermal mass material **130H**.

[0107] In operation, when a container **150** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) is inserted into the capsule **210H** (e.g. into the vessel portion **210H1** and lid portion **210H2**) and then inserted into the chamber **126H**, and the lid **L** closed over the container vessel **120H**, the thermally conductive contact(s) **210H7** will be placed in thermal communication (e.g., thermally contact, directly contact) with a cold-side heat sink of the cooling system **200G** (e.g., similar to the heat sink **210** in FIG. **4**) that is itself in thermal communication with one or more TECs (e.g., similar to TEC **220** in FIG. **4**), where the one or more TECs are operated to remove heat from (e.g., cool) the cold side heat sink, which in turn removes heat from (e.g., cools) the thermally conductive contact(s) **210H7**. The thermally conductive contact(s) **210H7** in turn remove heat from the cavity **210H8** to thereby cool the container **150**, as well as remove heat from the thermal mass material **130H** in the cavity **210H5** to thereby charge the thermal mass material **130H**. In one implementation, the cold side heat sink thermally contacts one of the thermally conductive contacts **210H7**. In another implementation, the cold side heat sink thermally contacts both of the thermally conductive contacts **210H7**. For example, the cold side heat sink in the lid **L** can thermally contact the thermally conductive contact **210H7** at one end of the capsule **210H** as well as thermally contact an inner wall of the chamber **126H** that itself contacts the thermally conductive contact **210H7** at the opposite end of the capsule **210H**.

[0108] The capsule **210H** can be removed along with the container **150** (e.g., one at a time, two at a time, etc.) from the container vessel **120H** (e.g., for placement in a user's purse, backpack, work bag during a commute or travel, etc.). Optionally, the capsule **210H** can maintain the container **150** in a cooled state for an extended period of time (e.g., between about 1 hour and about 15 hours, about 14 hours,

between about 1 hour and about 10 hours, between about 1 hour and about 3 hours, about 2 hours, etc.). The capsule 210H can maintain the container 150 approximately at a temperature of about 2-8 degrees Celsius. When the capsule 210H receives or houses the container 150 and is then inserted into the chamber 126H of the container vessel 120H, the capsule 210H can interface with the cooling system 200H and operate as a heat transfer interface between the cooling system 200H (e.g., between one or more TECs 220H of the cooling system 200H and the container 150) to help cool and/or heat the container 150. For example, when the cooling system 200H is used to cool the container 150, the capsule 210H can function as a heat sink to remove heat (e.g., cool) the container 150 that is disposed in the capsule 210H.

[0109] In one implementation, the cooling system 200H receives power via a power cord PC that can be connected to a wall outlet. However, the power cord PC can have other suitable connectors that allow the cooling system 200H to receive power from a power source other than a wall outlet. Power can be provided from the container vessel 120H, to which the power cord PC is connected, to the cooling system 200H in the lid via one or more electrical contacts on a rim of the container vessel 120H and on the lid L (e.g., similar to electrical contacts 282 described above in connection with FIG. 3). In another implementation, the power cord PC is excluded and the container vessel 120H can have one or more batteries (such as batteries 277 in FIG. 4) that provide power to the cooling system 200H (e.g., via electrical contacts, such as contacts 282 in FIG. 3) when the lid L is disposed over the container vessel 120H.

[0110] FIGS. 16B-16C show another implementation of the capsule 210H' for use with a container system 100H' and cooling system 200H'. The capsule 210H', container system 100H' and cooling system 200H' are similar to the capsule 210H, container system 100H and cooling system 200H described above in connection with FIGS. 16-16A. Thus, references numerals used to designate the various components of the capsule 210H, container system 100H and cooling system 200H are identical to those used for identifying the corresponding components of the capsule 210H', container system 100H' and cooling system 200H' in FIGS. 16B-16C, except that an "'" is added to the numerical identifier. Therefore, the structure and description for the various components of the capsule 210H, container system 100H and cooling system 200H in FIGS. 16-16A is understood to also apply to the corresponding components of the capsule 210H', container system 100H' and cooling system 200H' in FIGS. 16B-16C, except as described below.

[0111] The capsule 210H' differs from the capsule 210H in that the thermally conductive contact(s) 210H7 are excluded. The capsule 210H' has a movable mass 162H disposed in the cavity 210H9' between the intermediate wall 210H6' and the outer wall 210H3'. The movable mass 162H can optionally be a magnet. In another implementation, the movable mass 162H can be a metal block. The movable mass 162H can optionally be movably coupled to the intermediate wall 210H6' by a flexible thermally conductive element 164H, which operates as a thermal bridge between the movable mass 162H and the thermal mass material 130H'. In one implementation, the flexible thermally conductive element 164H can be made of copper. However, the flexible thermally conductive element 164H can be made of other suitable thermally conductive materials. The flexible

thermally conductive element 164H can be a leaf spring or similar resilient member that is attached at one end to the intermediate wall 210H6' and at its other end to the movable mass 162H. The movable mass 162H can optionally move within the second cavity 210H9' (e.g., a vacuum insulated cavity) between a first position where it is in contact with the intermediate wall 210H6' and a second position where it is in contact with the outer wall 210H3' of the capsule 210H'.

[0112] The container vessel 120H' can include one or more magnets 160H adjacent a wall of the chamber(s) 126H'. In one implementation, the one or more magnets 160H are permanent magnets. In another implementation, the one or more magnets 160H are electromagnets. The one or more magnets 160H can be in thermal communication with a cold side heat sink of the cooling system 200H' (e.g., via a wall or surface of the container vessel 120H', such as a wall of the chamber(s) 126H' that interfaces with the cold side heat sink when the lid L is placed on the container vessel 120H').

[0113] In operation, when a container 150 (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) is inserted into the capsule 210H' (e.g. into the vessel portion 210H1' and lid portion 210H2') and then inserted into the chamber 126H', and the lid L closed over the container vessel 120H', the one or more magnets 160H in the container vessel 120H' draw the movable mass 162H into contact with the outer wall 210H3' of the capsule 210H'. The cooling system 200H' draws heat out of the cavity 210H8' of the capsule 210H' (e.g., via operation of one or more TECs to draw heat from cold side heat sink, which itself draws heat from surfaces of components in the container vessel 120H' in thermal communication with the magnet 160H) by drawing heat from the thermal mass material 130H' via the flexible thermally conductive element 164H and contact between the movable mass 162H, outer wall 210H3' and magnet 160H. As heat is drawn from the thermal mass material 130H' to charge it, it also draws heat from the cavity 210H8' via the inner wall 210H4'. The magnet 160H and movable mass 162H (e.g., magnet, metallic component) therefore operate to form a thermal bridge through the cavity 210H9' (e.g., vacuum insulated cavity) to the thermal mass material 130H'.

[0114] The capsule 210H' can be removed along with the container 150 (e.g., one at a time, two at a time, etc.) from the container vessel 120H' (e.g., for placement in a user's purse, backpack, work bag during a commute or travel, etc.). Optionally, the capsule 210H' can maintain the container 150 in a cooled state for an extended period of time (e.g., between about 1 hour and about 15 hours, about 14 hours, between about 1 hour and about 10 hours, between about 1 hour and about 3 hours, about 2 hours, etc.). The capsule 210H' can maintain the container 150 approximately at a temperature of about 2-8 degrees Celsius.

[0115] The capsule(s) 210H, 210H' can optionally have a wireless transmitter and/or transceiver and a power source (e.g., battery) disposed therein (e.g., disposed in the cavity 210H9, 210H9'), and can have a temperature sensors in communication with the cavity 210H8, 210H8' (e.g., in thermal contact with the inner wall 210H4, 210H4'). The wireless transmitter and/or transceiver can optionally allow connectivity of the capsule(s) 210H, 210H' with an electronic device (e.g., a mobile electronic device, such as a smartphone), such as via an app on the electronic device, and can transmit sensed temperature information to the electronic device for tracking of internal temperature of the

capsule 210H', 210H. Optionally, the transmitter and/or transceiver can transmit an alert signal to the electronic device (e.g., visual alert, audible alert), such as a notification via the app, if the sensed temperature exceeds a temperature range (e.g., predetermined temperature range, preselected temperature limit) for the medication in the container 150. When the capsule 210H, 210H' is inserted into the chamber 126H, 126H' of the container vessel 120H, 120H', the transmitter and/or transceiver can also wirelessly transmit sensed temperature data sensed by the temperature sensor to the electronic device. Optionally, when in the container vessel 120H, 120H', the battery in the capsule(s) 210H, 210H' can be recharged (e.g., via induction power transfer, or via electrical contacts). In addition to maintaining the container 150 (and medication in the container 150) at or below a predetermined temperature range (e.g., 2-8 degrees C.) for a prolonged period of time (e.g., up to 14 hours, up to 10 hours, up to 5 hours, up to 3 hours, etc.), the capsule(s) 210H, 210H' can protect the container 150 therein from damage (e.g., breaking, spillage) if the capsule 210H, 210H' is dropped.

[0116] FIGS. 17-17B show a container system 100I (e.g., a capsule container) that includes a cooling system 200I. The container system 100I and cooling system 200I are similar to the container system 100H and cooling system 200H described above in connection with FIGS. 16-16A. Thus, references numerals used to designate the various components of the container system 100I and cooling system 200I are identical to those used for identifying the corresponding components of the container system 100H and cooling system 200H in FIGS. 16-16A, except that an "I" instead of an "H" is added to the numerical identifier. Therefore, the structure and description for the various components of the container system 100H and cooling system 200H in FIGS. 16-16A is understood to also apply to the corresponding components of the container system 100I and cooling system 200I in FIGS. 17-17B, except as described below.

[0117] As shown in FIG. 17, the container system 100I has a container vessel 120I and a lid L. The lid L can include a cooling system 200I. The container vessel 120I can optionally have one or more chambers 126I that extend to corresponding one or more openings 123I, each chamber 126I sized to receive and hold a container 150 (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.). Though FIG. 17 shows the container vessel 120I having six chambers 126I, one of skill in the art will recognize that the container vessel 120I can have more or fewer chambers 126I than shown in FIG. 17. Optionally, the container vessel 120I can have a chamber 126I2 that extends to an opening 123I2, the chamber 126I2 sized to receive a capsule 210I, which itself can hold one or more (e.g., one, two, etc.) containers 150 (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.), as further described below.

[0118] In one implementation, the container vessel 120I can have the same or similar structure as shown and described above for the container vessel 120, 120E, 120F, 120G, 120H. Optionally, the container vessel 120I can have a cavity between the chamber(s) 126I and the outer surface of the container vessel 120I that is vacuum insulated. In another implementation, the container vessel 120I excludes vacuum insulation and can instead have a gap or cavity between the chamber(s) 126I and an outer surface of the container vessel 120I that is filled with air. In still another

implementation, the container vessel 120I can have a gap or cavity between the chamber(s) 126I and an outer surface of the container vessel 120I that includes an insulating material.

[0119] FIGS. 17A-17B shows one implementation of a capsule 210I having a vessel portion 210I1 and a lid portion 210I2 (attached via a hinge 211I) that together can enclose one or more containers 150 (e.g., two containers 150 in FIG. 17A). The hinge 211I allows the lid portion 210I2 to be moved between a closed position an open position relative to the vessel portion 210I1. In the closed position, the lid portion 210I2 can optionally be held against the vessel portion 210I1 (e.g., by one or more magnetic surfaces of the lid portion 210I2 and/or vessel portion 210I1) to inhibit (e.g., prevent) the container 150 from inadvertently falling out of the capsule 210I.

[0120] The vessel portion 210I1 and lid portion 210I2 have an outer surface 210I3 and an inner surface 210I4 that defines a cavity 210I8 that receives the container(s) 150. The vessel portion 210I1 and lid portion 210I2 can also have an intermediate wall 210I6 radially between the inner surface 210I4 and the outer surface 210I3 that define a first cavity 210I5 between the inner wall 210I4 and the intermediate wall 210I6 and a second cavity 210I9 between the intermediate wall 210I6 and the outer surface 210I3. Optionally, the second cavity 210I5 can be vacuum insulated (i.e., the second cavity 210I5 can be under vacuum or negative pressure force). Optionally, the first cavity 210I5 can house a thermal mass material 130I. In one implementation, the thermal mass material 130I is a phase change material PCM (e.g., a solid-solid PCM, a solid-fluid PCM) that can transition from a heat absorbing state to a heat releasing state at a transition temperature. In another implementation, the cavity 210I5 is excluded and the capsule 210I instead has a wall that extends between the inner surface 210I4 and the intermediate wall(s) 210I6 that can absorb and release heat.

[0121] In operation, when a container 150 (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) is inserted into the capsule 210I (e.g. into the vessel portion 210I1) and then inserted into the chamber 126I, and the lid L closed over the container vessel 120I, the lid portion 210I2 can be in the open position relative to the vessel portion 210I1 (see FIGS. 17, 17A), allowing the thermal mass material 130I in the cavity 210I5 to be placed in thermal communication (e.g., thermally contact, directly contact) with a cold-side heat sink of the cooling system 200I (e.g., similar to the heat sink 210 in FIG. 4) that is itself in thermal communication with one or more TECs (e.g., similar to TEC 220 in FIG. 4), where the one or more TECs are operated to remove heat from (e.g., cool) the cold side heat sink, which in turn removes heat from (e.g., cools) the thermal mass material 130I and cavity 210I8 in the capsule 210I, as well as any containers 150 in the capsule 210I.

[0122] The capsule 210I can be removed along with one or more containers 150 (e.g., one at a time, two at a time, etc.) from the container vessel 120I (e.g., for placement in a user's purse, backpack, work bag during a commute or travel, etc.). Optionally, the capsule 210I can maintain the container(s) 150 in a cooled state for an extended period of time (e.g., between about 1 hour and about 15 hours, about 14 hours, between about 1 hour and about 10 hours, between about 1 hour and about 3 hours, about 2 hours, etc.). The

capsule **210I** can maintain the container **150** approximately at a temperature of about 2-8 degrees Celsius.

[0123] The capsule **210I** can optionally have a wireless transmitter and/or transceiver and a power source (e.g., battery) disposed therein (e.g., disposed in the cavity **210I9**), and can have a temperature sensors in communication with the cavity **210I8** (e.g., in thermal contact with the inner wall **210I4**). The wireless transmitter and/or transceiver can optionally allow connectivity of the capsule **210I** with an electronic device (e.g., a mobile electronic device, such as a smartphone), such as via an app on the electronic device, and can transmit sensed temperature information to the electronic device for tracking of internal temperature of the capsule **210I**. Optionally, the transmitter and/or transceiver can transmit an alert signal to the electronic device (e.g., visual alert, audible alert), such as a notification via the app, if the sensed temperature exceeds a temperature range (e.g., predetermined temperature range, preselected temperature limit) for the medication in the container **150**. When the capsule **210I** is inserted into the chamber **126I** of the container vessel **120I**, the transmitter and/or transceiver can also wirelessly transmit sensed temperature data sensed by the temperature sensor to the electronic device. Optionally, when in the container vessel **120I**, the battery in the capsule (s) **210I** can be recharged (e.g., via induction power transfer, or via electrical contacts). In addition to maintaining the container **150** (and medication in the container **150**) at or below a predetermined temperature range (e.g., 2-8 degrees C.) for a prolonged period of time (e.g., up to 14 hours, up to 10 hours, up to 5 hours, up to 3 hours, etc.), the capsule **210I** can protect the container **150** therein from damage (e.g., breaking, spillage) if the capsule **210I** is dropped.

[0124] In one implementation, the cooling system **200I** receives power via a power cord PC that can be connected to a wall outlet. However, the power cord PC can have other suitable connectors that allow the cooling system **200I** to receive power from a power source other than a wall outlet. Power can be provided from the container vessel **120I**, to which the power cord PC is connected, to the cooling system **200I** in the lid via one or more electrical contacts on a rim of the container vessel **120I** and on the lid L (e.g., similar to electrical contacts **282** described above in connection with FIG. 3). In another implementation, the power cord PC is excluded and the container vessel **120I** can have one or more batteries (such as batteries **277** in FIG. 4) that provide power to the cooling system **200I** (e.g., via electrical contacts, such as contacts **282** in FIG. 3) when the lid L is disposed over the container vessel **120I**.

[0125] FIGS. 18-18B show a container system **100J** (e.g., a cartridge container) that includes a cooling system **200J**. The container system **100J** and cooling system **200J** are similar to the container system **100H** and cooling system **200H** described above in connection with FIGS. 16-16A. Thus, references numerals used to designate the various components of the container system **100J** and cooling system **200J** are identical to those used for identifying the corresponding components of the container system **100H** and cooling system **200H** in FIGS. 16-16A, except that a "J" instead of an "H" is added to the numerical identifier. Therefore, the structure and description for the various components of the container system **100H** and cooling system **200H** in FIGS. 16-16A is understood to also apply to

the corresponding components of the container system **100J** and cooling system **200J** in FIGS. 18-18B, except as described below.

[0126] As shown in FIG. 18, the container system **100J** has a container vessel **120J** and a lid L. The lid L can include a cooling system **200J**. The container vessel **120J** can optionally have one or more chambers **126J** that extend to corresponding one or more openings **123J**, each chamber **126J** sized to receive and hold a container **150J** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.). In FIG. 16, the container **150J** is a cartridge that can be separately inserted into an injector device (e.g., injector pen) **170J** (see FIG. 18B), as discussed further below. The container vessel **120J** differs from the container vessel **120H** in that the opening(s) **123J** and chamber(s) **126J** are sized to receive container(s) **150J** that are cartridges. Though FIG. 18 shows the container vessel **120J** having six chambers **126J**, each being sized to removably receive a container **150J** (e.g., a cartridge), one of skill in the art will recognize that the container vessel **120J** can have more or fewer chambers **126J** than shown in FIG. 18.

[0127] In one implementation, the container vessel **120J** can have the same or similar structure as shown and described above for the container vessel **120**, **120E**, **120F**, **120G**, **120H**, **120I** and can maintain the container(s) **150** in a cooled state of approximately at a temperature of about 2-8 degrees Celsius. Optionally, the container vessel **120J** can have a cavity between the chamber(s) **126J** and the outer surface of the container vessel **120J** that is vacuum insulated. In another implementation, the container vessel **120J** excludes vacuum insulation and can instead have a gap or cavity between the chamber(s) **126J** and an outer surface of the container vessel **120J** that is filled with air. In still another implementation, the container vessel **120J** can have a gap or cavity between the chamber(s) **126J** and an outer surface of the container vessel **120J** that includes an insulating material.

[0128] FIG. 18A shows one implementation of a container **150J** (e.g. a cartridge, an injector pen) that can optionally house a medication (e.g., epinephrine, insulin, a vaccine, etc.). the container **150J** can have a temperature sensor **152J** and a radiofrequency identification (RFID) tag or chip **154J**, with the temperature sensors **152J** being in communication (e.g., electrically connected) with the RFID chip **154J**. The RFID chip **154J** can store temperature data sensed by the temperature sensor **152J**. Advantageously, the temperature sensor **152J** can track the temperature of the container **150J** from when it leaves the distribution center to when it arrives at a person's (consumer's) home, and to when it needs to be administered. The temperature data sensed by the temperature sensor **152J** is stored in the RFID chip **154J**, thereby providing a temperature history of the container **150J** from when it leaves the distribution center to when it arrives at a person's (consumer's) home, and to when it needs to be administered. In one implementation, the container vessel **120J** can have an optional RFID reader that can read the RFID chip **154J** once the container **150J** is inserted into the chamber **126J** of the container vessel **120J** to capture the temperature history stored in the RFID chip **154J**. Optionally, the container system **100J** can inform the user (e.g., via one or both of a graphical user interface on the container vessel **120J** and an app on an electronic device paired with the container system **100J**) that the medication in the con-

tainer 150J (e.g., cartridge) can be delivered (e.g., that the temperature history read from the RFID chip 154J indicates the medication in the container 150J has been maintained within a predetermined temperature range, so that the medication is deemed effective for delivery).

[0129] FIG. 18B shows an injection device 170J (e.g., auto injection device) into which the container 150J can be inserted prior to use (e.g., prior to application of the auto injection device on the user to deliver a medication in the container 150J, such as via a needle of the injection device 170J). When the container 150J (e.g., cartridge) is removed from the container vessel 120J and placed into the injection device 170J, an optional RFID reader in the injection device 170J can read the RFID chip 154J and send an alert to the user (via one or both of a graphical user interface on the injection device 170J and an app on an electronic device paired with the injection device 170J) that the medication can be delivered (e.g., that the temperature history read from the RFID chip 154J indicates the medication in the container 150 has been maintained within a predetermined temperature range, so that the medication is deemed effective for delivery).

[0130] In operation, when a container 150J (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, etc.) is inserted into the chamber 126J, and the lid L closed over the container vessel 120J, the container 150J can optionally be placed in thermal communication (e.g., thermally contact, directly contact) with a cold-side heat sink of the cooling system 200J (e.g., similar to the heat sink 210 in FIG. 4) that is itself in thermal communication with one or more TECs (e.g., similar to TEC 220 in FIG. 4), where the one or more TECs are operated to remove heat from (e.g., cool) the cold side heat sink, which in turn removes heat from (e.g., cools) the container(s) 150J in the vessel container 120J.

[0131] Optionally, the container 150J can optionally have a wireless transmitter and/or transceiver and a power source (e.g., battery) disposed therein. The wireless transmitter and/or transceiver can optionally allow connectivity of the container 150J with an electronic device (e.g., a mobile electronic device, such as a smartphone), such as via an app on the electronic device, and can transmit sensed temperature information (from the temperature sensor 152J) to the electronic device for tracking of internal temperature of the container 150J (e.g., in addition to or in place of tracking the sensed temperature history of the container 150J via the RFID chip 154J). Optionally, the transmitter and/or transceiver can transmit an alert signal to the electronic device (e.g., visual alert, audible alert), such as a notification via the app, if the sensed temperature exceeds a temperature range (e.g., predetermined temperature range, preselected temperature limit) for the medication in the container 150J. When the container 150J is inserted into the chamber 126J of the container vessel 120J, the transmitter and/or transceiver can also wirelessly transmit sensed temperature data sensed by the temperature sensor 152J to the electronic device. Optionally, when in the container vessel 120J, the battery in the container 150J can be recharged (e.g., via induction power transfer, or via electrical contacts).

[0132] In one implementation, the cooling system 200J receives power via a power cord PC that can be connected to a wall outlet. However, the power cord PC can have other suitable connectors that allow the cooling system 200J to receive power from a power source other than a wall outlet.

Power can be provided from the container vessel 120J, to which the power cord PC is connected, to the cooling system 200J in the lid via one or more electrical contacts on a rim of the container vessel 120J and on the lid L (e.g., similar to electrical contacts 282 described above in connection with FIG. 3). In another implementation, the power cord PC is excluded and the container vessel 120J can have one or more batteries (such as batteries 277 in FIG. 4) that provide power to the cooling system 200J (e.g., via electrical contacts, such as contacts 282 in FIG. 3) when the lid L is disposed over the container vessel 120J.

[0133] FIG. 19A shows a container system 100K (e.g., a medicine cooler container) that includes a cooling system 200K. Though the container system 100K has a generally box shape, in other implementations it can have a generally cylindrical or tube shape, similar to the container system 100, 100E, 100F, 100G, 100H, 100I, 100J. In one implementation, the cooling system 200K can be in the lid L of the container system 100K and can be similar to (e.g., have the same or similar components as) the cooling system 200, 200E, 200F, 200G, 200H, 200I, 200J. In another implementation, the cooling system can be disposed in a portion of the container vessel 120K (e.g. a bottom portion of the container vessel 120K).

[0134] As shown in FIG. 19A, the container system 100K can include a display screen 180K. Though FIG. 19A shows the display screen 180K on the lid L, it can alternatively (or additionally) be incorporated into a side surface 122K of the container vessel 120K. The display screen 180K can be an electronic ink or E-ink display (e.g., electrophoretic ink display). In another implementation, the display screen 180K can be a digital display (e.g., liquid crystal display or LCD, light emitting diode or LED, etc.). Optionally, the display screen 180K can display a label 182K (e.g., a shipping label with one or more of an address of sender, an address of recipient, a Maxi Code machine readable symbol, a QR code, a routing code, a barcode, and a tracking number). The container system 100K can also include a user interface 184K. In FIG. 19A, the user interface 184K is a button on the lid L. In another implementation, the user interface 184K is disposed on the side surface 122K of the container vessel 120K. In one implementation, the user interface 184K is a depressible button. In another implementation, the user interface 184K is a capacitive sensor (e.g., touch sensitive sensor). In another implementation, the user interface 184K is a sliding switch (e.g., sliding lever). In another implementation, the user interface 184K is a rotatable dial. Advantageously, actuation of the user interface 184K can alter the information shown on the display screen 180K, such as the form of a shipping label shown on an E-ink display screen 180K. For example, actuation of the user interface 184K, can switch the text associated with the sender and receiver, allowing the container system 100K to be shipped back to the sender once the receiving party is done with it.

[0135] FIG. 19B shows a block diagram of electronics 500 of the container system 100K. The electronics 500 can include circuitry EM' (e.g., including one or more processors on a printed circuit board). The circuitry EM' communicate with one or more batteries PS', with the display screen 180K, and with the user interface 184K. Optionally, a memory module 185K is in communication with the circuitry EM'. In one implementation, the memory module 185K can optionally be disposed on the same printed circuit board as other

components of the circuitry EM'. The circuitry EM' optionally controls the information displayed on the display screen 180K. Information (e.g., sender address, recipient address, etc.) can be communicated to the circuitry EM' via an input module 186K. The input module 186K can receive such information wirelessly (e.g., via radiofrequency or RF communication, via infrared or IR communication, via WiFi 802.11, via BLUETOOTH®, etc.), such as using a wand (e.g., a radiofrequency or RF wand that is waved over the container system 100K, such as over the display screen 180K, where the wand is connected to a computer system where the shipping information is contained). Once received by the input module 186K, the information (e.g., shipping information for a shipping label) to be displayed on the display screen 180K can be electronically saved in the memory module 185K. Advantageously, the one or more batteries PS' can power the electronics 500, and therefore the display screen 180K for a plurality of uses of the container system 100K (e.g., during shipping of the container system 100K up to one-thousand times).

[0136] FIG. 20A shows a block diagram of one method 700A for shipping the container system 100K. At step 710, one or more containers, such as containers 150, 150J (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, vaccines, medicine such as insulin, epinephrine, etc.) are placed in the container vessel 120K of the container system 100K, such as at a distribution facility for the containers 150, 150J. At step 720, the lid L is closed over the container vessel 120K once finished loading all containers 150, 150J into the container vessel 120K. Optionally, the lid L is locked to the container vessel 120K (e.g., via a magnetically actuated lock, including an electromagnet actuated when the lid is closed that can be turned off with a code, such as a digital code). At step 730, information (e.g., shipping label information) is communicated to the container system 100K. For example, as discussed above, a radiofrequency (RF) wand can be waved over the container system 100K (e.g., over the lid L) to transfer the shipping information to the input module 186K of the electronics 500 of the container system 100K. At step 740, the container system 100K is shipped to the recipient (e.g., displayed on the shipping label 182K on the display screen 180K).

[0137] FIG. 20B shows a block diagram of a method 700B for returning the container system 100K. At step 750, after receiving the container system 100K, the lid L can be opened relative to the container vessel 120K. Optionally, prior to opening the lid L, the lid L is unlocked relative to the container vessel 100K (e.g., using a code, such as a digital code, provided to the recipient from the shipper). At step 760, the one or more containers 150, 150J are removed from the container vessel 120K. At step 770, the lid L is closed over the container vessel 120K. At step 780, the user interface 184K (e.g., button) is actuated to switch the information of the sender and recipient in the display screen 180 with each other, advantageously allowing the return of the container system 100K to the original sender to be used again without having to reenter shipping information on the display screen 180K. The display screen 180K and label 182K advantageously facilitate the shipping of the container system 100K without having to print any separate labels for the container system 100K. Further, the display screen 180K and user interface 184K advantageously facilitate return of the container system 100K to the sender (e.g. without having to reenter

shipping information, without having to print any labels), where the container system 100K can be reused to ship containers 150, 150J (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, vaccines, medicine such as insulin, epinephrine, etc.) again, such as to the same or a different recipient. The reuse of the container system 100K for delivery of perishable material (e.g., medicine) advantageously reduces the cost of shipping by allowing the reuse of the container vessel 120K (e.g., as compared to commonly used cardboard containers, which are disposed of after one use).

[0138] FIGS. 21A-21D show different screens of a graphical user interface (GUI) used on a remote electronic device (e.g., mobile electronic device, such as a mobile phone, tablet computer). The GUI advantageously allows a user to interface with the cooling system 200, 200E, 200F, 200G, 200H, 200I, 200J, 200K, 200L provide control settings (e.g., temperature presets for different medications in the containers 150, 150J), provide scheduling information (e.g., for the consumption of medication in the containers 150, 150J), provide alerts (e.g., battery life of the cooling system, temperature of the container(s) 150, 150J). The GUI can provide additional information not shown on the screens in FIGS. 21A-21D. Via the GUI, a user can communicate with the cooling system 200, 200E, 200F, 200G, 200H, 200I, 200J, 200K, 200L when they are ready to ingest the contents of the container 150, 150J and the system 200, 200E, 200F, 200G, 200H, 200I, 200J, 200K, 200L can optionally heat one of the containers 150, 150J a predetermined temperature (e.g., body temperature, room temperature) and optionally alert the user when ready (via the GUI) to notify the user when the contents (e.g., medication) is ready for consumption. Optionally, where the container vessel 120, 120E, 120F, 120G, 120H, 120I, 120J, 120K, 120L includes more than one container 150, 150J, the user can communicate via the GUI with the system 200, 200E, 200F, 200G, 200H, 200I, 200J, 200K, 200L to prepare (e.g., heat) one of the containers (e.g., to body temperature) while the rest of the containers 150, 150J in the container vessel 100 remain in a cooled state. Optionally, once the container 150, 150J has been prepared (e.g., heated), in addition to notifying the user that the contents (e.g., medication) in the container 150, 150J is ready for consumption, it can also actuate the chamber 126, 126', 126E, 126F1, 126F2, 126G1, 126L to move it to the extended position (e.g., via one of the linear actuation mechanisms disclosed herein) so when the user removes the lid from the container vessel 120, 120E, 120F, 120G, 120H, 120I, 120J, 120K, 120L the user can readily identify which of the containers 150, 150J is the one that is ready for consumption (e.g., which one has been heated to room temperature or body temperature), while the rest of the chambers 126, 126', 126E, 126F1, 126F2, 126G1, 126L remain in the retracted position.

[0139] FIGS. 22A-22B show a container system 100L (e.g., capsule container) that includes a cooling system 200L. Some of the features of the container system 100L and cooling system 200L are similar to features of the container system 100-100K and cooling system 200-200K in FIGS. 1-19A. Thus, reference numerals used to designate the various components of the container system 100L and cooling system 200K are identical to those used for identifying the corresponding components of the container system 100-100K and cooling system 200-200K in FIGS. 1-19A, except that an "L" has been added to the numerical identifier.

Therefore, the structure and description for the various features of the container system 100-100K and cooling system 200-200K and how it's operated and controlled in FIGS. 1-19A are understood to apply to the corresponding features of the container system 100L and cooling system 200L in FIGS. 22A-22B, except as described below.

[0140] The container system 100L has a container vessel 120L that is optionally cylindrical. The container vessel 120L is optionally a cooler with active temperature control provided by the cooling system 200L to cool the contents of the container vessel 120L and/or maintain the contents of the vessel 120L in a cooled or chilled state. Optionally, the vessel 120L can hold therein one or more (e.g., a plurality of) separate containers 150 (e.g., medicine containers, such as injector pens, vials, cartridges (such as for injector pens), etc.). Optionally, the one or more (e.g., plurality of) separate containers 150 that can be inserted into the container vessel 120L can contain a medication or medicine (e.g., epinephrine, insulin, vaccines, etc.).

[0141] The container vessel 120L has an outer wall 121L that extends between a proximal end 122L that has an opening and a distal end 124L having a base 125L. The opening is selectively closed by a lid L removably attached to the proximal end 122L. The vessel 120L has an inner wall 126AL and a base wall 126BL that together define an open chamber 126L that can receive and hold contents to be cooled therein (e.g., medicine containers, such as one or more vials, cartridges, injector pens, etc.). The vessel 120L can optionally have an intermediate wall 126CL spaced about the inner wall 126AL and base wall 126BL, such that the intermediate wall 126CL is at least partially disposed between the outer wall 121L and the inner wall 126AL. The intermediate wall 126CL is spaced apart from the inner wall 126AL and base wall 126BL so as to define a gap 128L between the intermediate wall 126CL and the inner wall 126AL and base wall 126BL. The gap 128L can optionally be under vacuum so that the inner wall 126AL and base wall 126BL are vacuum insulated relative to the intermediate wall 126CL and the outer wall 121L of the vessel 120L.

[0142] Optionally, one or more of the inner wall 126AL, intermediate wall 126CL and outer wall 121L can be made of metal (e.g., stainless steel). In one implementation, the inner wall 126AL, base wall 126BL and intermediate wall 126CL are made of metal (e.g., stainless steel). In another implementation, one or more portions (e.g., outer wall 121L, intermediate wall 126CL and/or inner wall 126AL) of the vessel 120L can be made of plastic.

[0143] The vessel 120L has a cavity 127L between the base wall 126BL and the base 125L of the vessel 120L. The cavity 127L can optionally house electronics, such as, for example, one or more batteries 277L and one or more printed circuit boards (PCBA) with circuitry that controls the operation of the cooling system 200L. In one implementation, the cavity 127L can optionally house a power button or switch actuatable by a user through the bottom of the vessel 200L. Optionally, at least a portion of the base 125L (e.g. a cap of the base 125L) is removable to access the electronics in the cavity 127L (e.g., to replace the one or more batteries 277L, perform maintenance on the electronics, such as the PCBA, etc.). The power button or switch is accessible by a user (e.g., can be pressed to turn on the cooling system 200L, pressed to turn off the cooling system 200L, pressed to pair the cooling system 200L with a mobile

electronic device, etc.). Optionally, the power switch can be located generally at the center of the base 125L.

[0144] The cooling system 200L is optionally at least partially housed in the vessel 120L. In one implementation, the cooling system 200L can include a first heat sink (cold side heat sink) 210L in thermal communication with one or more thermoelectric elements (TECs) 220L, such as Peltier element(s), and can be in thermal communication with the chamber 126L of the vessel 120L (e.g., via contact with the inner wall 126AL, via conduction with air in the chamber 126L, etc.). The first heat sink 210L portion outside the vessel 120L communicates with the first heat sink 210L portion inside the vessel 120L via a first heat sink 210L portion (e.g., bridge portion) that interconnects the portions of the first heat sink 210L outside and inside the vessel 120L.

[0145] The one or more TECs 220L are selectively operated (e.g., by the circuitry) to draw heat from the first heat sink (e.g., cold-side heat sink) 210L and transfer it to the second heat sink (hot-side heat sink) 230L. A fan 280L is selectively operable to draw air into the vessel 120L (e.g., into a channel FP of the vessel 120L) to dissipate heat from the second heat sink 230L, thereby allowing the TECs 220L to draw further heat from the first heat sink 210L, and thereby draw heat from the chamber 126L. During operation of the fan 280L, intake air flow F_i is drawn through one or more intake vents 203L (having one or more openings) in the vessel 120L and over the second heat sink 230L (where the air flow removes heat from the second heat sink 230L), after which the exhaust air flow F_o flows out of one or more exhaust vents 205L (having one or more openings) in the vessel 120L.

[0146] The chamber 126L optionally receives and holds one or more (e.g., a plurality of) containers 150 (e.g., medicine containers, such as injector pens or cartridges for injector pens, vials, etc.). In one implementation, the first heat sink 210L can be made of aluminum. However, the first heat sink 210L can be made of other suitable materials (e.g., metals with high thermal conductivity).

[0147] The electronics (e.g., PCBA, batteries 277L) can electrically communicate with the fan 280L and TECs 220L. Accordingly, power can be provided from the batteries 277L to the TECs 220L and/or fan 280L, and the circuitry (e.g., in or on the PCBA) can control the operation of the TECs 220L and/or fan 280L.

[0148] The container 100L can optionally have a visual display on the outer wall 121L of the vessel 120L (e.g., on the lid L). The visual display can optionally display one or more of the temperature in the chamber 126L, the temperature of the first heat sink 210L, the ambient temperature, a charge level or percentage for the one or more batteries 277L, and amount of time left before recharging of the batteries 277L is needed, etc. The visual display can optionally include a user interface (e.g., pressure sensitive buttons, capacitance touch buttons, etc.) to adjust (up or down) the temperature preset at which the cooling system 200L is to cool the chamber 126L. Accordingly, the operation of the container 100L (e.g., of the cooling system 200L) can be selected via the visual display and user interface on a surface of the container 100L. Optionally, the visual display can include one or more hidden-til-lit LEDs. Optionally, the visual display can include an electrophoretic or electronic ink (e-ink) display. In one variation, the container 100L can optionally include a hidden-til-lit LED that can selectively illuminate (e.g., to indicate one or more operating functions

of the container **100L**, such as to indicate that the cooling system **200L** is in operation). The LED can optionally be a multi-color LED selectively operable to indicate one or more operating conditions of the container **100L** (e.g., green if normal operation, red if abnormal operation, such as low battery charge or inadequate cooling for sensed ambient temperature, etc.).

[0149] In operation, the cooling system **200L** can optionally be actuated by pressing a power button. Optionally, the cooling system **200L** can additionally (or alternatively) be actuated remotely (e.g., wirelessly) via a remote electronic device, such as a mobile phone, tablet computer, laptop computer, etc. that wirelessly communicates with the cooling system **200L** (e.g., with a receiver or transceiver of the circuitry). In still another implementation, the cooling system **200L** can automatically cool the chamber **126L** when the lid **L** is in a closed position on the vessel **120L**. The chamber **126L** can be cooled to a predetermined and/or a user selected temperature or temperature range, or automatically cooled to a temperature preset corresponding to the contents in the containers **150** (e.g., insulin, epinephrine, vaccines, etc.). The user selected temperature or temperature range can be selected via a user interface on the container **100L** and/or via the remote electronic device.

[0150] In one variation, the container system **100L** is powered using 12 VDC power (e.g., from one or more batteries **277L** or a power base on which the vessel **120L** is placed). In another variation, the container system **100L** is powered using 120 VAC or 240 VAC power, for example using a power base. The circuitry in the container **100L** can include a surge protector to inhibit damage to the electronics in the container **100L** from a power surge. The container system **100L** is advantageously easy to assemble and simpler to use. For example, inclusion of the cooling system **200** in the vessel **120L** makes it easier for users with limitations in hand articulation (e.g., users suffering from arthritis) to open the lid **L** (e.g., because it is lighter or weighs less) to remove the container(s) **150** (e.g., vaccines, insulin, medical containers) from the chamber **126L**. The lid **L** can optionally be insulated (e.g., be made of a hollow plastic body filled with foam insulation, such as light density Styrofoam).

[0151] While certain embodiments of the inventions have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms. For example, though the features disclosed herein are in described for medicine containers, the features are applicable to containers that are not medicine containers (e.g., portable coolers for food, chilled water cooler/bottle, etc.) and the invention is understood to extend to such other containers. Furthermore, various omissions, substitutions and changes in the systems and methods described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure. Accordingly, the scope of the present inventions is defined only by reference to the appended claims.

[0152] Features, materials, characteristics, or groups described in conjunction with a particular aspect, embodiment, or example are to be understood to be applicable to any other aspect, embodiment or example described in this section or elsewhere in this specification unless incompat-

ible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The protection is not restricted to the details of any foregoing embodiments. The protection extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

[0153] Furthermore, certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as a subcombination or variation of a subcombination.

[0154] Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, or that all operations be performed, to achieve desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Those skilled in the art will appreciate that in some embodiments, the actual steps taken in the processes illustrated and/or disclosed may differ from those shown in the figures. Depending on the embodiment, certain of the steps described above may be removed, others may be added. Furthermore, the features and attributes of the specific embodiments disclosed above may be combined in different ways to form additional embodiments, all of which fall within the scope of the present disclosure. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products.

[0155] For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. Not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

[0156] Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not

generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or steps are included or are to be performed in any particular embodiment.

[0157] Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

[0158] Language of degree used herein, such as the terms “approximately,” “about,” “generally,” and “substantially” as used herein represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “approximately,” “about,” “generally,” and “substantially” may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the stated amount. As another example, in certain embodiments, the terms “generally parallel” and “substantially parallel” refer to a value, amount, or characteristic that departs from exactly parallel by less than or equal to 15 degrees, 10 degrees, 5 degrees, 3 degrees, 1 degree, or 0.1 degree.

[0159] The scope of the present disclosure is not intended to be limited by the specific disclosures of preferred embodiments in this section or elsewhere in this specification, and may be defined by claims as presented in this section or elsewhere in this specification or as presented in the future. The language of the claims is to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive.

What is claimed is:

1. A portable cooler container with active temperature control for one or more injector pens, comprising:

a container vessel having an outer wall that extends between a proximal end that has an opening and a distal end having a base, the opening selectively closed by a lid, the container vessel having a chamber sized to receive one or more injector pens, a chamber wall being spaced inward of an inner wall to define a gap between the chamber wall and the inner wall, the gap filled with a phase change material; and

a temperature control system comprising:

one or more thermoelectric elements configured to heat or cool at least a portion of the chamber,

a heat sink in thermal communication with one side of the one or more thermoelectric elements and in thermal communication with the phase change material,

one or more power storage elements, and

circuitry configured to control the one or more thermoelectric elements to heat or cool at least the portion of the chamber to a predetermined temperature or temperature range,

wherein the one or more thermoelectric elements are operable to heat or cool the phase change material.

2. The container of claim 1, wherein a second heat sink is in thermal communication with an opposite side of the one or more thermoelectric elements.

3. The container of claim 1, further comprising a user interface configured to display information indicative of a temperature in the chamber, an ambient temperature, or a charge level of the one or more power storage elements.

4. The container of claim 1, wherein the chamber comprises two spaced apart chambers, each chamber configured to receive one injector pen.

5. The container of claim 2, wherein one or more fans are operable to draw air through one or more air intake vents, to flow said air over the second heat sink to dissipate heat from the second heat sink, and to then flow said air through one or more exhaust vents.

6. The container of claim 1, wherein the chamber wall is cylindrical.

7. The container of claim 1, further comprising a power base configured to receive the container vessel thereon and to provide power to the one or more thermoelectric elements or one or more power storage elements via inductive coupling.

8. A portable cooler container with active temperature control for one or more injector pens, comprising:

a container vessel having an outer wall that extends between a proximal end that has an opening and a distal end having a base, the opening selectively closed by a lid, the container vessel having a chamber sized to receive one or more injector pens, a chamber wall being spaced inward of an inner wall to define a gap between the chamber wall and the inner wall, the gap filled with a phase change material; and

a temperature control system comprising:

one or more thermoelectric elements configured to heat or cool at least a portion of the chamber,

a heat sink in thermal communication with one side of the one or more thermoelectric elements and in thermal communication with the phase change material,

one or more power storage elements,

circuitry configured to control the one or more thermoelectric elements to heat or cool at least the portion of the chamber to a predetermined temperature or temperature range; and

one or more sensors configured to sense one or more parameters of the chamber and to communicate the sensed parameters to the circuitry,

wherein the one or more thermoelectric elements are operable to heat or cool the phase change material.

9. The container of claim 8, wherein a second heat sink is in thermal communication with an opposite side of the one or more thermoelectric elements.

10. The container of claim 8, further comprising a user interface configured to display information indicative of a temperature in the chamber, an ambient temperature, or a charge level of the one or more power storage elements.

11. The container of claim 8, wherein the chamber comprises two spaced apart chambers, each chamber configured to receive one injector pen.

12. The container of claim 8, wherein the chamber wall is cylindrical.

13. The container of claim 8, further comprising a power base configured to receive the container vessel thereon and

to provide power to the one or more thermoelectric elements or one or more power storage elements via inductive coupling.

14. A portable cooler container with active temperature control for one or more injector pens, comprising:

a container vessel having an outer wall that extends between a proximal end that has an opening and a distal end having a base, the opening selectively closed by a lid, the container vessel having a chamber sized to receive one or more injector pens, a chamber wall being spaced inward of an inner wall to define a gap between the chamber wall and the inner wall, the gap filled with a phase change material; and

a temperature control system comprising:

one or more thermoelectric elements,

a heat sink in thermal communication with one side of the one or more thermoelectric elements and in thermal communication with the phase change material,

one or more power storage elements,

circuitry configured to control the one or more thermoelectric elements; and

one or more sensors configured to sense one or more parameters of the chamber and to communicate the sensed parameters to the circuitry,

wherein the one or more thermoelectric elements are operable to heat or cool the phase change material.

15. The container of claim **14**, further comprising a user interface configured to display information indicative of a temperature in the chamber, an ambient temperature, or a charge level of the one or more power storage elements.

16. The container of claim **14**, wherein the chamber comprises two spaced apart chambers, each chamber configured to receive one injector pen.

17. The container of claim **14**, further comprising one or more light emitting diodes (LEDs) operable to selectively illuminate to indicate one or more operating functions of the container.

18. The container of claim **14**, wherein the chamber wall is cylindrical.

19. The container of claim **14**, further comprising a power base configured to receive the container vessel thereon and to provide power to the one or more thermoelectric elements or one or more power storage elements via inductive coupling.

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